Unburdened Fields, Mobilized Workers: Labor Market Responses to Agricultural Tax Reform and Intergenerational Consequences

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Abstract

In many developing economies, the coexistence of low productivity agriculture and surplus labor in farming reflects a fundamental sectoral misallocation that slows structural transformation and limits long run growth. This paper studies how the agricultural tax reform in China between 2000 and 2006 reshaped this distorted equilibrium. First, by employing a staggered difference-in-differences (DID) design, I document a large and immediate financial benefits from tax reform on rural families. I argue that rural households responded by investing heavily in agricultural machinery, which substituted for manual labor and reduced labor input in farm sectors. These adjustments facilitated labor reallocation into higher-productivity nonfarm employment. Second, I employ a cohort DID specification to study intergenerational impacts. I show the exposure to tax reform at school ages significantly raised the educational attainments of teenagers and generated more long-run benefits in their adulthood, suggesting a persistent gain in human capital accumulation and labor allocation adjustment. Finally, I show these changes are different between genders and are concentrated on females and girls. The results highlight how fiscal reforms that relax production constraints can trigger technology adoption, mitigate labor misallocation, and generate persistent improvements in human capital formation.

Keywords: agricultural tax reform; mechanization; labor reallocation; intergenerational human capital.

JEL Codes: J43, O15, O18, Q18.

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1 Introduction

Large and persistent productivity differences across countries remain one of the central puzzles in development economics. Less developed economies typically exhibit substantially lower aggregate productivity than developed economies, and a growing body of research attributes this gap to the fact that poorer countries allocate a disproportionately large share of their labor force to low-productivity agricultural sectors in order to satisfy subsistence requirements (Restuccia et al., 2008; Lagakos and Waugh, 2013; Young, 2013; Adamopoulos and Restuccia, 2014; Gollin et al., 2014). This excessive concentration of labor in farming induces a fundamental sectoral misallocation that depresses agricultural productivity and constrains the movement of workers into higher-productivity non-agricultural sectors. These distortions slow structural transformation and limit long-run growth (Restuccia and Rogerson, 2017; Chari et al., 2021; Adamopoulos et al., 2022).

In this study, I leverage the agricultural tax reform in China to examine how public financial policy can promote labor reallocation toward nonfarm sectors through technology adoption and improvements in agricultural productivity. Implemented in the early 2000s, the reform abolished the long-standing agricultural tax and a wide range of local surcharges historically imposed on rural households. By removing these agriculture-related levies, the reform substantially lowered the effective cost of agricultural production. Existing studies show that the reform generated sizable gains in agricultural productivity and contributed to a structural shift of labor out of agriculture and into non-agricultural sectors (Yu and Jensen, 2010; Xu et al., 2012; Li et al., 2024). In particular, Li et al. (2024) emphasizes the role of farm capital investment in driving these gains. Building on this insight, I argue that the reform-induced improvement in households' budget environment created strong incentives to upgrade production technology. The greater investment in agricultural machinery could substitute for manual farm labor, reduce labor demand in agricultural production, and thereby release workers into higher-productivity nonfarm activities. These arguments are broadly consistent with a growing literature linking technology adoption in agriculture to productivity growth and labor reallocation across sectors in other developing environment (Bustos et al., 2016; McArthur and McCord, 2017; Ivanic and Martin, 2018; Shamdasani, 2021; Bustos et al., 2020; Gollin et al., 2021).

I examine these dynamics by exploiting the county-level rollout of the agricultural tax reform between 2000 and 2006 in a staggered difference-in-differences (DID) design. Using household data from the National Fixed Point Survey (NFPS) and China Health and Retirement Longitudinal Study (CHARLS) Life History, I find that the reform increased grain output and pre-tax farm income per unit of farmland by 37.9 and 49.2 percent, respectively, and household nonfarm income per capita by 69.3 percent. These gains were accompanied by greater investment in energized farm machinery, together with a 21.1 percent decline in household farm labor input and a 53.4 percent increase in nonfarm labor input.

Furthermore, I examine whether this labor reallocation differs by gender. Existing studies and government reports show that rural women are more likely to remain in farm production or household responsibilities, due to childcare, marriage, and traditional gender norms (Mu and van de Walle, 2011; Magnani and Zhu, 2012; de Brauw and Giles, 2018; Wu et al., 2021), which suggests that labor-saving mechanization may release female labor from agriculture more strongly than male labor. Consistent with this view, I find that the reform reduced full-time farm among women by about four times as much as among men. This gender difference is larger in regions with more male-biased sex ratios at birth, where stronger son preference may have constrained women's pre-reform labor allocation.

These production-side adjustments may also affect the young generation by reducing the opportunity cost of schooling and relaxing household budget constraints. Consistent with evidence that agricultural labor demand can crowd out children's education (Albertus et al., 2020; Xu, 2021; Hu et al., 2024), the reform should improve human capital accumulation and long-run outcomes. As the decline in agricultural labor is larger among women, these effects may be particularly pronounced for girls.

To estimate these intergenerational effects, I employ a cohort DID design that compares teenagers aged 12–15 at the time of the local reform with slightly older cohorts aged 16–19. The cutoff at age 15 reflects both the end of compulsory schooling and the typical age of entry into high school in China. I find that exposure to the reform increased girls' high school enrollment and completion by 1.7 and 1.5 percentage points, respectively, while the corresponding estimates for boys are small and statistically insignificant. This finding aligns with Hu et al. (2024), which documents a decline in teenagers educational attainment when the Household Responsibility System raised the labor demand in agriculture. Beyond schooling, I also document that girls exposed before high school entry were also less likely to receive government poverty assistance and less likely to be out of the labor force due to housework in adulthood.

I examine the mechanisms of these intergenerational gender differences. The gains are larger for girls with siblings. This finding supports the view that the reform generated an income effect by directly relaxing household budget constraints. I also find larger gains for girls whose mothers never worked in nonfarm sectors and in regions with more male-biased sex ratios at birth. These two measures proxy for more traditional gender norms and a more rigid gender division of labor within the household and local environment. The larger effects suggest that the reform improved girls' outcomes partly by reducing this opportunity cost. These findings on gender differences in mechanization and educational attainment are closely related to a growing literature in China (Wu et al., 2021; Lin et al., 2021; Xie et al., 2022; Glewwe et al., 2022; Chen and Zhang, 2025) and other developing settings (Afridi et al., 2023; Bacher, 2024; Doss and Gottlieb, 2025; Euler et al., 2025).

This work makes three contributions to the literature. First, it contributes to the literature on low labor productivity and sectoral misallocation in less developed economies. This paper provides micro-level causal evidence on how policy-induced mechanization can raise agricultural productivity and facilitate the reallocation of labor out of agriculture. It complements the macro-development literature on sectoral misallocation by offering a household-level perspective on the mechanisms through which agricultural distortions shape productivity and labor allocation (Hsieh and Klenow, 2009; Lagakos and Waugh, 2013; Adamopoulos and Restuccia, 2014; Gollin et al., 2014; Jovanovic, 2014; Restuccia and Rogerson, 2017; Herrendorf and Schoellman, 2018; Bryan and Morten, 2019; Baqaee and Farhi, 2020; Chari et al., 2021; Adamopoulos et al., 2022; Porzio et al., 2022; Rossi, 2022).

Second, the paper contributes to the literature on technology adoption in labor-intensive agricultural systems. A large literature examines how mechanization and other forms of capital adoption affect agricultural productivity (Bustos et al., 2016; Ivanic and Martin, 2018; Meng et al., 2024a), labor supply (Benin, 2015; McArthur and McCord, 2017; Shamdasani, 2021; Afridi et al., 2023; Ma et al., 2024), and structural transformation (Bustos et al., 2020; Gollin et al., 2021; Acemoglu et al., 2024) in developing economies. This paper contributes to that literature by showing how a fiscal reform can shape household machinery adoption and labor allocation through changes in the effective cost of agricultural production.

Third, the paper adds to the growing body of research on gender gaps in the labor market and in its transition to young generations, particularly in the context of low-income and agrarian economies (Bick and Fuchs-Schündeln, 2018; Field et al., 2021; Wu et al., 2021; Barr et al., 2022; Borella et al., 2023; Aizer et al., 2024; Bacher, 2024; Abou Daher et al., 2025; Boelmann et al., 2025). In particular, this work is tightly related with a few discussions on gender heterogeneity in the adoption of advanced technology and mechanization in agricultural sectors (Takeshima, 2024; Chen and Zhang, 2025; Euler et al., 2025). The paper underscores how productivity-enhancing reforms can alleviate gender-specific constraints and contribute to narrowing gender gaps in less developed settings.

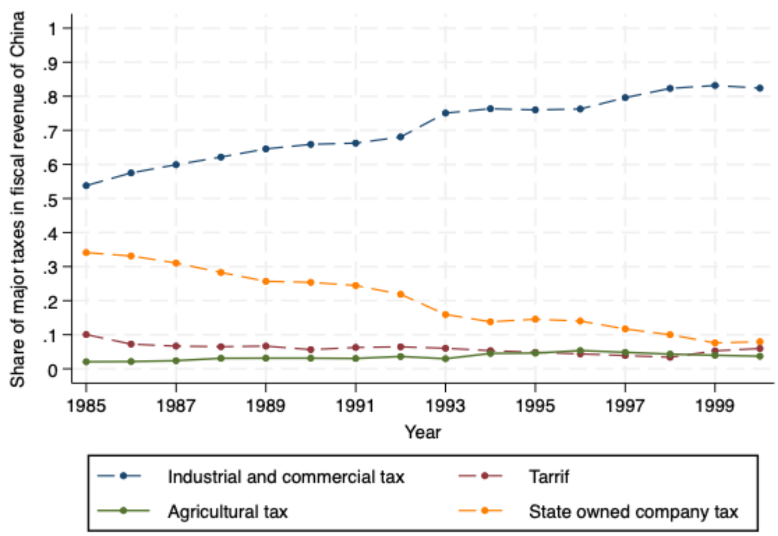
The remainder of the paper is organized as follows. Section 2 provides background on the agricultural tax reform, rural labor allocation, and the schooling system in China. Section 3 describes the data and empirical strategies. Sections 4 and 5 present the short-run effects on household production and labor allocation and the intergenerational effects on children, respectively. Section 6 presents the robustness checks. Section 7 concludes.

2 Background and Conceptualization

2.1 Agricultural Tax Reform in China

The Chinese government initiated the agricultural tax reform in the early 2000s to abolish the long-standing agricultural tax system. Originating in the imperial era, the agricultural tax had long been an important source of local fiscal revenue. By the late 1990s, however, it had become increasingly incompatible with China's rural development goals. As shown in the first panel of Figure 1, the agricultural tax absorbed more than 15 percent of household farming income before the reform, imposing a heavy burden on rural families (Kung, 1995). At the same time, its contribution to total government fiscal revenue had fallen to only about 5 percent by the 1990s. Against this backdrop, the central government initiated a tax reform in 2000, later expanded it nationwide, and ultimately abolished the agricultural tax (Yu and Jensen, 2010).

Before the reform, agriculture-related taxation in China consisted mainly of three components: (i) the agricultural tax, (ii) the agricultural surcharge, and (iii) the tax on agricultural specialty products. The agricultural tax was levied on the output of grain crops and several cash crops,



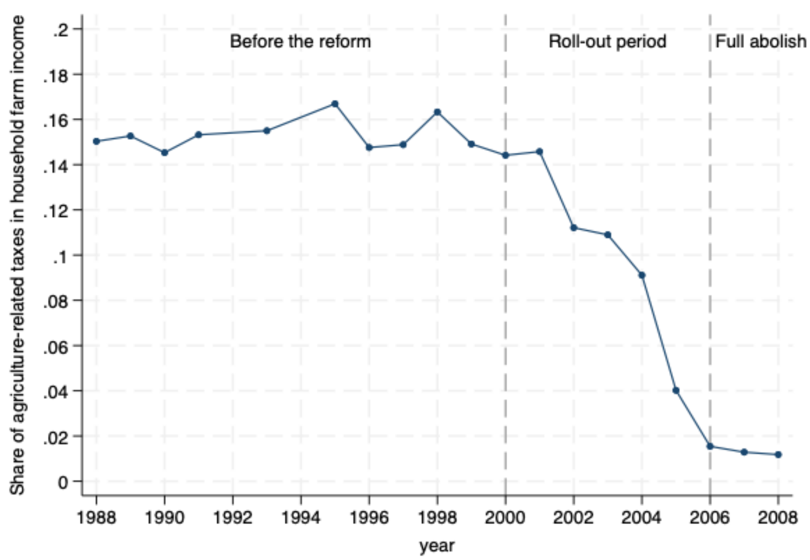


Figure 1: Tax Revenue and Agriculture-related Tax Burden Before the Tax Reform

Note: Tax revenue data is from annual statistic books. Tax share is from NFPS.

such as cotton. Its statutory rate was 15.5 percent at the national level, although the exact rate varied slightly across regions. The agricultural surcharge was an additional levy attached to the agricultural tax and was primarily collected by local governments to finance local expenditures. Its rate typically ranged from 10 to 20 percent of the agricultural tax liability. The tax on agricultural specialty products applied to other commercial agricultural products, such as fruits and tea. Tax rates differed across products and were generally higher than the agricultural tax rate. In addition to these three components, rural households were also subject to various miscellaneous taxes and fees, which were mainly used to finance local spending on education and infrastructure.

In 2000, the reform was first piloted in Anhui Province, with a small number of counties in other provinces also authorized to participate. After a one-year suspension in 2001 for policy evaluation, the reform resumed in 2002 and continued to expand through 2003, with additional counties and provinces joining each year. By the end of this stage, the reform had expanded to counties in 20 provinces. During this initial stage, the agricultural tax and the agricultural surcharge were consolidated into a single tax category, with the combined rate capped at 7 percent of crop income. Miscellaneous taxes and fees were also abolished and replaced by central government fiscal transfers.

In 2004, the central government announced a nationwide roll-out of the reform. At the same time, the upper limit of the agricultural tax rate was gradually reduced to 4 to 6 percent, and the tax rate on agricultural specialty products was lowered accordingly. Finally in 2006, all of these taxes, except the tax on tobacco, were fully abolished. As shown in the second panel of Figure 1, the reform substantially reduced the tax burden on rural households in the post-reform years.

2.2 Rural Employment and Agricultural Mechanization

Rural households in China are characterized by the coexistence of small-scale family farming and nonfarm wage employment within the same household. Since the Household Responsibility System (HRS) in the late 1970s, land has been contracted to individual households, which make their own production decisions but typically work on small plots with limited capital (Deininger et al., 2014; Chen and Lan, 2020; Adamopoulos et al., 2022). Agricultural production remains highly labor-intensive and relies primarily on family labor, while hired farm labor and formal labor markets for agricultural work are rare (Sheng et al., 2019). At the same time, large and persistent rural-urban wage gaps have encouraged substantial temporary and seasonal migration into construction, manufacturing, and services in urban areas. As a result, many rural households allocate some members to nonfarm work while others remain engaged in farming and household responsibilities (Fan and Stark, 2008; Bosker et al., 2012).

This household structure is closely related to the broader problem of sectoral misallocation. In particular, land and capital are not systematically allocated toward more productive farms, and too many workers remain concentrated in the agricultural sector.[1] This misallocation depresses productivity in agriculture and slows the reallocation of labor toward higher-productivity non-agricultural activities.

I argue that the agricultural tax reform in the early 2000s may have altered this equilibrium through agricultural mechanization. Li et al. (2024) suggests that, once relieved of heavy taxation, households were more able to allocate resources toward productive capital, which can improve agricultural productivity, substitute for manual labor, and facilitate labor reallocation across sectors.[2]

Another important feature of nonfarm work in rural China is its pronounced gender imbalance. Men account for about 70 percent of migrant workers, while women account for only about 30 percent. This imbalance in nonfarm migrant employment implies that women are disproportionately represented among those who remain in agricultural production or household responsibilities. Existing studies point to two possible explanations. First, nonfarm employment opportunities available to rural workers have been disproportionately concentrated in physically demanding sectors such as construction and manufacturing, where men may have a comparative advantage (Shauman, 2010). Second, traditional gender norms may make women more likely to remain in villages or at home to take on farming tasks and household duties (Magnani and Zhu, 2012; de Brauw and Giles, 2018; Wu et al., 2021). Against this background, I argue that mechanization is likely to generate gender-differentiated effects, with a stronger tendency to release female labor from agricultural work.[3]

[1] Although temporary migrant work has become increasingly common among rural households, the imbalance in labor allocation across sectors remains substantial. See Chari et al. (2021) and Adamopoulos et al. (2022).

[2] In Appendix Figure A1, I provide descriptive evidence on this argument, where households in the post-reform period exhibit higher levels of agricultural machinery holdings and capital value, accompanied by lower manual farm labor input measured by labor input days.

[3] Appendix Figure A2 provides descriptive evidence consistent with this interpretation by showing that the decline in agricultural labor is more pronounced among women in the post-reform period.

2.3 Schooling and Educational Decisions

China's schooling system before college consists of six years of primary school, three years of middle school, and three years of high school. The first nine years have been compulsory since the enactment of the Compulsory Education Law in 1986 and typically cover children between the ages of 6 and 15 (must complete primary and middle schools). For rural families, the end of compulsory schooling often marks a critical decision point. Once compulsory schooling ends, children may either continue to high school by paying tuition[4] or leave school to assist with household farming or local employment. This institutional threshold makes high school entry a natural margin for educational decisions and the central focus of my analysis in this study.[5]

I argue two channels through which the reform may have affected high school attainment. One possible channel is an income effect, as tax relief directly increases household resources available for children's education (Hotz et al., 2023). Another possibility is that the reform changed the relative value of continued schooling within rural households. As agricultural production became less labor-intensive and nonfarm opportunities expanded, the opportunity cost of schooling may have fallen while its perceived returns may have risen. This interpretation is consistent with evidence from the literature. For example, Hu et al. (2024) show that exposure to the Household Responsibility System reform in China increased the return to agricultural labor and reduced school enrollment, and Nakamura et al. (2022) find that parents shifted educational investment toward skills better aligned with labor demand after forced relocation.

3 Data and Empirical Strategy

3.1 Data Collection and Variables

This study relies on two broad categories of data. The first consists of information on the agricultural tax reform itself, including county-level implementation timing and corresponding tax rates. The second consists of micro-level data, which I use to examine how the reform affected rural households' production decisions, labor allocation, and children's long-run outcomes.

[4] China also has vocational high schools, which focus more on vocational skills than academic training and also require tuition. In this paper, I refer to both academic and vocational high schools as high school.

[5] Appendix Figure A3 shows that cohorts exposed to the reform before high school entry exhibit higher high school completion rates. This descriptive change suggests that the reform may have had a positive effect on high school attainment.

Agricultural tax reform records. I compile county-level information on the agricultural tax reform, including implementation years and corresponding tax rates, using local fiscal gazetteers and historical policy documents issued by local governments. I begin by reviewing local fiscal gazetteers, which typically record the year in which the reform was implemented province-wide as well as the corresponding province-level tax rates, but generally do not document the pilot counties that adopted the reform earlier. I then use historical government documents to supplement this information by (i) filling in provinces missing from the fiscal gazetteers and (ii) identifying pilot counties that implemented the reform before the province-wide expansion, together with their locations, implementation years, and applicable tax rates.

The second category of data consists of micro-level survey data, which are drawn from three datasets:

National Fixed Point Survey (household data). The National Fixed Point Survey (NFPS) is one of China's longest-running nationally representative rural panel surveys. It is conducted annually by the Research Center for Rural Economy under the Ministry of Agriculture and Rural Affairs of China and follows sampled rural households across provinces over time, providing detailed information at the household and village levels. This panel dataset has been widely used in empirical studies (Chari et al., 2021; Liu et al., 2023). In this study, I use household-level NFPS data from 1996 to 2008. These data provide the main household-level variables used in the short-run analysis. In particular, I use measures of agricultural output and household income to estimate the reform's effects on agricultural and non-agricultural gains, measures of farm asset and machinery values to estimate its effects on production capital investment, and measures of household labor input days and labor shares to estimate its effects on household labor allocation.

China Health and Retirement Longitudinal Study Life History (individual worker data). The China Health and Retirement Longitudinal Study (CHARLS) is a nationally representative panel survey launched in 2011 that follows Chinese residents aged 45 and above. In 2014, CHARLS conducted a special life history survey that collected retrospective information on respondents' employment histories up to 2014. The survey records the start and end years of each past job and classifies primary occupations into several categories. The quality of this survey has been verified in studies (Hu et al., 2024). Excluding work periods with military service, I use this information to construct an individual-level measure of whether a respondent's primary activity in a given year was full-time farm.[6] For this study, I use individuals' work histories from 1996 to 2008 and restrict the sample to respondents with rural registration. Because CHARLS provides individual-level employment histories, it allows me to estimate the reform's effects separately for men and women and to examine gender differences in labor reallocation. In addition, I use several household-related demographic variables in CHARLS to conduct heterogeneity analysis.

2010 Census Sample Survey (school-age teenager data). The 2010 Census Sample Survey is a one-year cross-sectional household survey conducted following China's Sixth National Population Census. It sampled 1% population in China and provides information on household demographics, housing, and place of residence, as well as individual characteristics such as gender, birth year, educational attainment, work status, and health conditions. Similar to CHARLS, I restrict the sample to individuals with rural registration. I use this dataset to examine high school attainment and other long-run outcomes. For this purpose, I restrict the sample to individuals who were between 12 and 19 years old in the year of the reform. This age range is chosen to ensure that all cohorts were exposed to the enforcement of Compulsory Education Law in 1986 and that high school attainment can be observed by 2010.[7]

To estimate the reform's effect on educational outcomes, a concern with using age 15 as the cutoff is that some cohorts in the control group may still be partially affected by the reform. One type consists of teenagers who deferred high school entry, which is not uncommon in rural China (Qihui, 2022; Chen, 2021; Chen and Park, 2021). For these individuals, the relevant cutoff would be above age 15 because they entered high school later than the standard age. Another concern is that the reform may also affect individuals who were already enrolled in high school when it occurred, for example by lowering dropout and thereby increasing the likelihood of completion. Both cases raise the possibility that the baseline cutoff may underestimate the effects on high school attainment.

To address this concern, I construct indicators for high school enrollment and high school

[6] Because CHARLS does not publicly reports respondents' county of residence, the empirical strategy based on the county-level rollout is not valid with this dataset. To address this limitation, I drop the sample years in which the reform had been implemented only in a subset of counties within a province, and define the post-treatment period as the years when the reform had been rolled out province-wide. See Section 3.2 for details.

[7] The upper bound is chosen so that all teenagers would have reached the primary school entry age of 6 in 1986 or later, which requires birth year 1980 or later. Since the reform began as early as 2000 in some areas, this implies an age of at most 20 in the reform year; I use 19 as a conservative bound. The lower bound is chosen so that high school attainment can be observed in 2010. Since high school completion is typically observed by age 18, individuals must have been born no later than 1992. Given that the latest reform implementation in this sample occurred in 2003, this implies an age of at least 11 in the reform year; I use 12 as a conservative bound.

completion as two measures of educational attainment. In constructing the high school completion measure, I exclude individuals who were still enrolled in high school in 2010. I also conduct additional robustness checks related to this concern in Section 6 by dropping partially affected control cohorts and showing that the reform had limited effects on high school dropout.

For the evaluation of other long-run outcomes potentially associated with improved educational attainment, I use receipt of government poverty-related assistance and engagement in full-time housework in 2010 as two additional outcome variables.

Besides these three datasets, I also use the 2000 Fifth Population Census, the 1996 First Agricultural Census, and the Chinese Household Income Project (CHIP) to construct variables for the heterogeneity analysis. Descriptive statistics are reported in Appendix Table A1.

3.2 Empirical Strategy

I employ three empirical approaches to estimate the treatment effects. First, I use a staggered DID specification to estimate the reform's effects on household outcomes and adult workers' labor allocation. Second, I use a cohort DID design to estimate the reform's effects on children's educational attainment and other long-run outcomes. Finally, I use event-study specifications to present the dynamic treatment effects and examine the parallel-trend assumption.

I estimate household and adult workers' responses to the tax reform using the following staggered DID specification:

$$y_{ict} = \beta \times reform_{ict} + \delta_i + \delta_t + \delta_c \times t + \varepsilon_{it} \tag{1}$$

where y_{ict} denotes outcomes related to household tax burden and income, agricultural capital investment, and household members' labor allocation. δ_i and δ_t denote household or individual fixed effects and year fixed effects, respectively. I also include county-specific linear time trends, captured by $\delta_c \times t$, to account for unobservable factors that change differentially across counties over time.

The treatment variable $reform_{ict}$ is a binary indicator equal to one if the county of residence c of household or individual i had implemented the reform by year t (regardless of the magnitude of the tax rate reduction), and zero otherwise. From the perspective of the policy's historical implementation, a natural design would define $reform_{ict}$ as the county-year reduction in the agricultural tax rate relative to its pre-reform baseline. The treatment variable in this way captures reform intensity and identifies marginal effects of tax cuts. However, this approach has been challenged by the recent DID literature (de Chaisemartin et al., 2024; Callaway et al., 2024), particularly when treatment intensity varies over time within the same county, which complicates identification and makes the TWFE estimate more difficult to interpret.

In this sense, I adopt the binary treatment definition so that, under the parallel trends assumption, the treatment parameter β identifies the *average treatment effect among treated observations*, regardless of the exact treatment intensity.[8] This specification is conservative in the sense that it gives up identification of marginal effects with respect to treatment intensity, but in return yields a more aggregated and more interpretable estimate of the reform's overall effect. Under this setup, the remaining identification issue is the negative-weight problem in staggered DID designs, I address which using an alternative estimator proposed by Callaway and Sant'Anna (2021) in Section 6.

The specification (1) relies on the county of residence of household or individual i. This information is available in the NFPS but not publicly disclosed in CHARLS, which reports only the respondent's city of residence. To make the empirical design applicable to CHARLS, I exclude the transitional years (generally one or two years) in which the reform had been implemented only in a subset of counties within a province, and define the post-treatment period as the years from province-wide implementation onward. This adjustment is unlikely to generate substantial bias since before province-wide implementation each province typically had fewer than ten pilot counties adopting the reform early, and in most cases fewer than five. It is unlikely that the overall treatment effect is driven primarily by these pilot counties. Formally, the DID specification applied to CHARLS is:

$$y_{ipt} = \beta \times reform_{ipt} + \delta_i + \delta_t + \delta_c \times t + \varepsilon_{ipt} \tag{2}$$

where $reform_{ipt}$ is an indicator equal to one if the reform had been rolled out province-wide in province p by year t, and zero otherwise. t excludes the years in which the reform had been

[8] Following Callaway et al. (2024), abstracting from staggered adoption, a binary-treatment DID identifies $ATT^o = E[ATT(d) \mid d > 0]$, where d denotes treatment intensity. If one abstracts from time-varying treatment intensity, a continuous-treatment DID identifies $\int_{d_L}^{d_U} w(l) ATT(l \mid l), dl$, where $ATT(l \mid l)$ is the average treatment effect of intensity l relative to no treatment for units with treatment intensity l, and the weighting function $w(l)$ can be negative for low values of l. With both staggered adoption and time-varying treatment intensity, the interpretation becomes substantially more complicated.

implemented only in a subset of counties within a province.[9]

Second, I use a cohort DID design (Duflo, 2001) to estimate the reform's effects on teenagers:

$$y_{icb} = \beta \times reform_{icb} + \delta_c + \delta_b + \delta_c \times b + \varepsilon_{ib} \tag{3}$$

where y_{icb} includes variables such as high school attainment and other long-run outcomes. $reform_{icb}$ is an indicator that equals to one if individual i was younger or equal 15 years old at the reform start year in county c, and zero otherwise. δ_c and δ_b denote county fixed effects and birth-year fixed effects, respectively. I include county-specific linear cohort trends, captured by $\delta_c \times b$.

Finally, I estimate the dynamic treatment effects using the following event-study specification to examine the pre-treatment parallel trends assumption:

$$y_{ict} = \sum_{k \neq -1} \beta_k \times D^k_{ict} + \delta_i + \delta_t + \delta_c \times t + \varepsilon_{it} \tag{4}$$

where y_{ict} follows the same outcomes as in the staggered DID specification in equation (1). D^k_{ict} is an indicator equal to one if county c is k years relative to the reform year in year t, and zero otherwise. $k = -1$ is omitted as the base group. For the CHARLS specification in equation (2), the corresponding event-study specification is defined at the province level so that the county subscript c is replaced by the province subscript p. And D^k_{ict} is an indicator equal to one if year t is k periods relative to the first year of province-wide reform implementation in province p, with the transitional years excluded.

For cohort DID,

$$y_{icb} = \sum_{k \neq 16} \beta_k \times D^k_{icb} + \delta_c + \delta_b + \delta_c \times b + \varepsilon_{ib} \tag{5}$$

where y_{icb} follows the same outcomes as in the cohort DID specification in equation (3). D^k_{icb} is an indicator equal to one if individual i was age k in the reform year in county c. $k = 12, 13, \cdots, 19$. $k = 16$ is omitted as the base group.

[9] CHARLS provides a masked identifier for the county of residence, which allows to include county-specific time trends in the specification.

4 Effects of Tax Reform on Households and Workers

In this section, I start by presenting the immediate benefits of tax deduction on farming income. I next show how rural families respond to the financial gains by reallocating resources in farming input. Then I show how this reallocation is different between genders and regions with different characteristics. Finally, I conclude this section by justifying the role of farming machinery in the above effects.

4.1 Main Results

Table 1 reports the estimated treatment effects of the agricultural tax reform on household production and income outcomes. I first examine the effects within the agricultural sector. The estimates in Column (1) and (2) show that the reform increased grain output per unit of household farmland by 37.9 percent and pre-tax farm income per unit of farmland by 49.2 percent. These estimates indicate that the reform substantially improved both agricultural output and the economic returns to land. The gains were not limited to the agricultural sector. In Column (3), the estimates show that the reform increased household nonfarm income per capita by 69.3 percent, which is a surprisingly large increase but consistent with the predictions of the conceptual framework.

Table 1: Agricultural and Non-Agricultural Gains

Outcome variable:	Grain produced per farmland (log) (1)	Pre-tax farm income per farmland (log) (2)	Nonfarm income per capita (log) (3)
$Reform_{it}$	0.379**	0.492***	0.693***
	(0.145)	(0.120)	(0.236)
Household FE	Y	Y	Y
Year FE	Y	Y	Y
County-specific trend	Y	Y	Y
Observation	25,407	25,407	25,407
R-squared	0.490	0.571	0.514

Note: This table uses NFPS household data and employs specification (1). $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

Corresponding to the three specifications in Table 1, Figure 2 presents the estimates of the event-study specification. The insignificant estimates before the reform years indicate the pre-treatment parallel trends. The estimates after the reform years show very similar dynamic treatment effects in both sign and magnitude, which supports the robustness of DID estimates, especially with the concern of the negative weights for TWFE-DID estimates in the staggered setting (Goodman-Bacon, 2021).

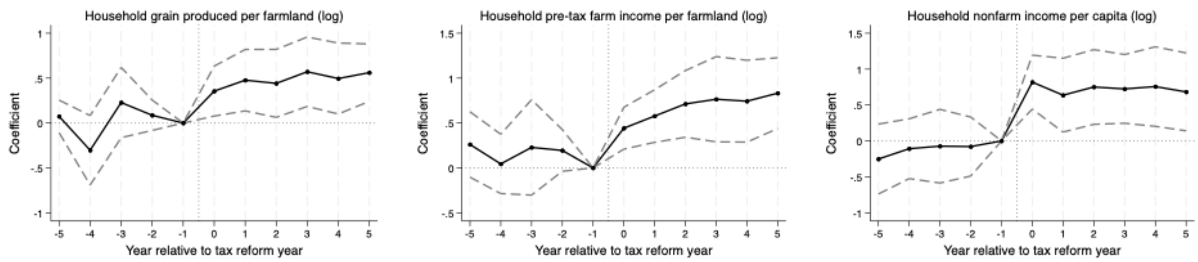


Figure 2: Event Study of Table 1

Note: This figure plots the coefficients and 95% confidence intervals of the event-study specification using data from Table 1.

To better understand the underlying margins of adjustment, I examine how the reform affected household production inputs. Table 2 reports the effects on agricultural production capital investment. In Column (1), the reform increased household total farm-related asset value per unit of farmland by 46.2 percent, which is very close to the findings documented in Li et al. (2024). This increase, however, was not evenly distributed across asset types. The hand-operated farm tool value per farmland in Column (2) was increased only 0.4 percent and not statistically significant. By contrast, the increase is concentrated in energized farm machinery. In Column (3) to (5), farm production machinery value per farmland rose by 56.8 percent, farm processing machinery value per farmland by 12.8 percent, and farm transport equipment value per farmland by 54.1 percent. The estimates suggest that households appear to have shifted investment toward more automated and mechanized agricultural equipment, while reducing reliance on more labor-intensive production tools. I plot the estimated coefficients of the corresponding event-study specifications in Figure 3, where the pre-treatment parallel trend is indicated by the insignificant estimates before the reform years. The DID estimates in Table 2 is also supported by the similar dynamic effects in the post-treatment periods.

This change in the composition of investment is accompanied by a reallocation of household labor across sectors. As shown in Table 3, the reform reduced household total farm labor input days per year by 21.1 percent, increased household nonfarm days per year by 53.4 percent, and lowered the share of household members engaged in full-time farm activities by 6.5 percentage

Table 2: Effects of Agricultural Tax Reform on Agricultural Production Assets

Outcome variable:	Total farm asset value per farmland (log) (1)	Hand-operated farm tool value per farmland (log) (2)	Farm production machinery value per farmland (log) (3)	Farm processing machinery value per farmland (log) (4)	Farm transport equipment value per farmland (log) (5)
$Reform_{it}$	0.462***	0.004	0.568***	0.128**	0.541***
	(0.097)	(0.062)	(0.170)	(0.061)	(0.082)
Household FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y
Observation	25,407	25,407	25,407	25,407	25,407
R-squared	0.809	0.817	0.805	0.743	0.680

Note: This table uses NFPS household data and employs specification (1). $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

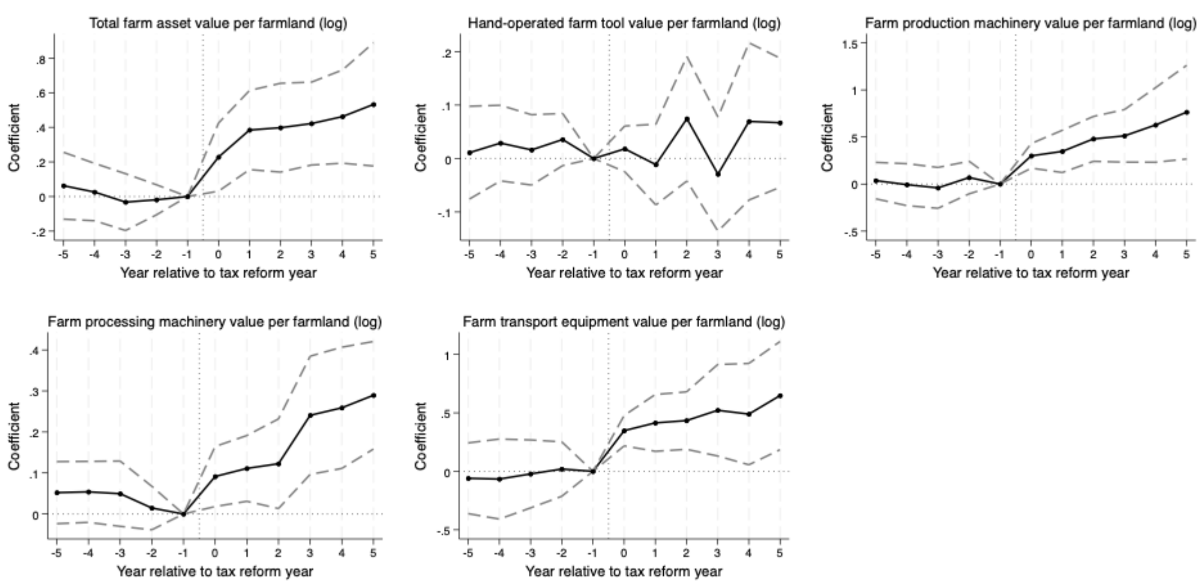


Figure 3: Event Study of Table 2

Note: This figure plots the coefficients and 95% confidence intervals of the event-study specification using data from Table 2.

points, which are generally consistent with but slightly higher than the estimated effects in Li et al. (2024). These estimates indicate that, following the reform, households reallocated labor away from farm production and toward nonfarm activities. Combined with the results on capital investment in Table 2, the evidence suggests that households appear to have used the tax benefits generated by the reform to invest in mechanized farm machinery that substituted for labor input in farm production, consequently releasing more labor resources into nonfarm work.

Table 3: Effects of Agricultural Tax Reform on Labor Allocation

Outcome variable:	Household farm labor input days (log) (1)	Household nonfarm labor input days (log) (2)	Household full-time farm labor share (3)
$Reform_{it}$	-0.211***	0.534**	-0.065***
	(0.056)	(0.208)	(0.012)
Household FE	Y	Y	Y
Year FE	Y	Y	Y
County-specific trend	Y	Y	Y
Observation	25,407	25,407	25,407
R-squared	0.558	0.491	0.573

Note: Column (1) to (3) use NFPS household data. This table employs specification (1). $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

This interpretation is also consistent with the increases in income in both the farm and nonfarm sectors reported in Table 1. For farm production, a large body of literature has documented that mechanization and the adoption of new technologies in agriculture can substantially raise productivity and returns (Bustos et al., 2016; McArthur and McCord, 2017; Meng et al., 2024a). For nonfarm activities, earnings are largely labor-driven in the context of China, as rural non-agricultural employment has been concentrated primarily in urban construction and labor-intensive manufacturing and service industries. In this context, the findings in this paper highlight the broader role of agricultural mechanization in improving both farm production and nonfarm employment by raising farm productivity and facilitating the reallocation of labor toward nonfarm activities. Figure 4 plots the estimated coefficients from the event-study specifications corresponding to Table 3. The insignificant estimates before the reform support the parallel-trends assumption.

Within these arguments, an important underlying assumption is the substitution (instead of complement) of farm machinery for manual labor in the farm production activities, which is ambiguous in the literature (Wang et al., 2016; Hamilton et al., 2022). To justify this assumption of

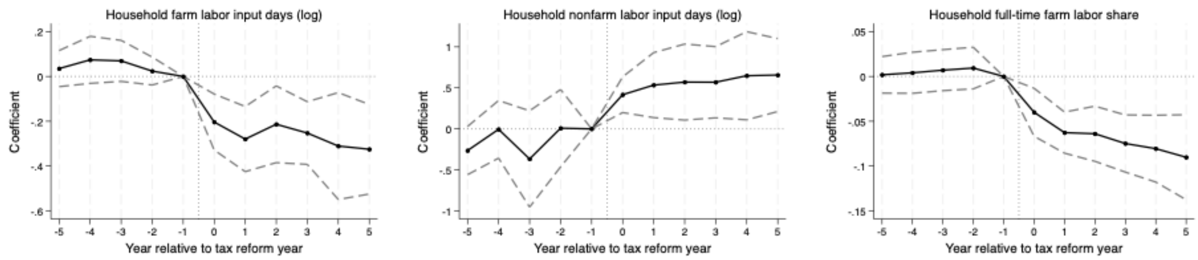


Figure 4: Event Study of Table 3

Note: This figure plots the coefficients and 95% confidence intervals of the event-study specification using data from Table 3.

the substitution relationship, I regress the likelihood of household workers engaged in full-time farm activities on the value of total farm asset. As shown in Figure 5, I find a negative correlation for the whole sample after controlling for the two-way fixed effects and the county-specific time trend using either pre-treatment and post-treatment samples. These results give suggestive evidence on the substitution between manual labor and the adoption of machinery.

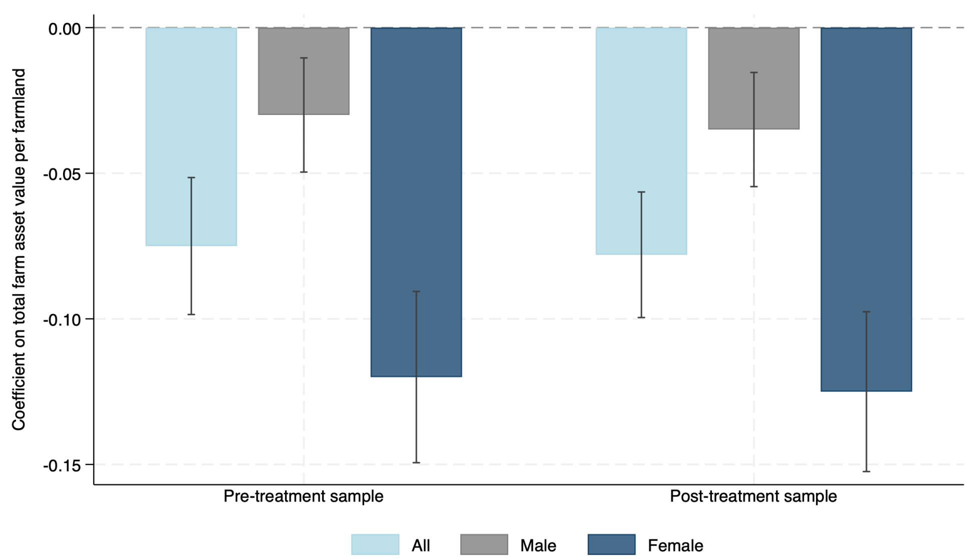


Figure 5: Regression of Full-Time Farm on Total Farm Asset Value per Farmland

Note: This figure plots the regression of full-time farm on the province-level average value of family total farm asset by controlling for the two-way fixed effects and the county-specific time trend. full-time farm is measured using the CHARLS Life History Survey and matched to province-level average family total farm asset value constructed from the NFPS.

4.2 Mechanism Analysis

The evidence above suggests that the reform increased household investment in mechanized farm equipment and reallocated labor away from agricultural production and toward non-agricultural activities. In this section, I examine whether this labor adjustment was indeed driven by mechanization. Table 4 shows heterogeneous treatment effects by interacting the reform indicator with proxies for the suitability of mechanized production and with direct measures of local mechanization levels.

Table 4: Mechanism Analysis of Mechanization and Labor Allocation

Interaction variable:	County average farmland ruggedness	Farmland consolidation	Mechanization level Baseline	Mechanization level Dynamic
	(1)	(2)	(3)	(4)
Panel A outcome variable: household farm labor input days (log)				
$Reform_{it}$	-0.405***	-0.156***	-0.188***	-0.181***
	(0.061)	(0.054)	(0.045)	(0.043)
$Reform_{it} \times interaction$	0.0005**	-0.011**	0.006**	-0.001***
	(0.0002)	(0.005)	(0.003)	(0.000)
R-squared	0.565	0.560	0.561	0.564
Panel B outcome variable: household nonfarm labor input days (log)				
$Reform_{it}$	1.806***	0.065	0.498**	0.477**
	(0.289)	(0.212)	(0.193)	(0.195)
$Reform_{it} \times interaction$	-0.003**	0.097***	-0.014**	0.001***
	(0.000)	(0.001)	(0.007)	(0.000)
R-squared	0.505	0.496	0.489	0.492
Panel C outcome variable: household full-time farm labor share				
$Reform_{it}$	-0.090***	-0.050***	-0.059***	-0.064***
	(0.015)	(0.012)	(0.012)	(0.012)
$Reform_{it} \times interaction$	0.0001***	-0.003***	0.001**	-0.001**
	(0.0000)	(0.000)	(0.000)	(0.000)
R-squared	0.575	0.575	0.570	0.573
Household FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y
Observation	25,407	25,407	25,407	25,407

Note: This table uses NFPS household data and repeats Table 3 Column (1) to (3) with interaction terms. Panel A to C use NFPS household data. The variables at the top are used as the interaction terms. Column (1) uses the average of farmland terrain ruggedness index of each county from Davis et al. (2025). Column (2) uses the average land area per farmland plot of each household to measure the farmland consolidation. Column (3) and (4) use the household pre-treatment average farm machinery value and the contemporaneous farm machinery value in each year, respectively. The interaction variables in Column (1) to (3) are measured in pre-treatment periods and thus time-invariant. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

In the first two columns, I exploit farmland ruggedness and farmland fragmentation levels to measure the suitability of mechanized production. Existing literature suggests the farm machinery more effectively improves farm productivity on farmland on flatter terrains (Ji et al., 2012; Afridi et al., 2023; Meng et al., 2024b) or on larger and less fragmented plots (Shen et al., 2025; Wang et al., 2026). In this sense, if mechanization is an important margin of adjustment, the labor-reallocating effect of the reform should be stronger in areas where machinery can be adopted more easily (i.e., areas with lower ruggedness or fragmentation levels).

Column (1) uses the ruggedness measure as the interaction variable. The ruggedness is derived from Davis et al. (2025), which is defined on the county level by the standard deviation of within-county elevations. In Panel A, the coefficient on the interaction term is 0.0005[10], indicating that the reform-induced decline in household farm labor input days becomes smaller in more rugged counties. In Panel B, the corresponding interaction coefficient is -0.003, showing that the increase in household nonfarm labor input days is also weaker in more rugged counties. In Panel C, the interaction coefficient is 0.0001, which implies that the reduction in the household full-time farm labor share is attenuated in counties with more rugged terrain. These estimates are consistent with the argument that rough terrain limits the effective use of machinery, thereby weakening the extent to which households can substitute capital for labor and reallocate workers out of agriculture.

The findings are similar in Column (2), where the interaction variable is the average area per household farmland plot, constructed so that a larger value corresponds to a lower degree of land fragmentation and greater suitability for machinery adoption. The estimates are consistent with the argument that the treatment effects are weaker in areas with higher degree of land fragmentation. Specifically, across the three panels, the interaction coefficients are -0.011 for household farm labor input days, 0.097 for household nonfarm labor input days, and -0.003 for the household full-time farm labor share.

More evidence comes from heterogeneity by local mechanization levels. Column (3) interacts the reform indicator with the baseline mechanization level. The interaction coefficient is 0.006 for household farm labor input days, -0.014 for household nonfarm labor input days, and 0.001 for household full-time farm labor share. These estimates indicate that the reform had smaller effects on labor reallocation in counties with higher pre-reform mechanization levels. In places where

[10] The interaction coefficient is numerically small because the ruggedness variable has a large raw scale. Since I focus on the sign and statistical significance rather than the magnitude or interpretation of a one-unit change, I keep the original measure without rescaling for transparency.

mechanization was less prevalent before the reform, households had greater scope to translate tax relief into additional machinery investment, as a larger share of rural labor remained concentrated in agricultural production. As a result, the labor-saving effect of machinery could work more strongly in these areas, allowing households to substitute capital for farm labor more effectively and reallocate more labor resources into non-agricultural activities.

Column (4) shows a similar implication where I use the dynamic mechanization level. This specification captures the relationship between contemporaneous household mechanization levels and labor allocation within the same period, which closely matches the idea illustrated in Figure 5. The interaction coefficient is -0.001 when the outcome is household farm labor input days, 0.001 when the outcome is household nonfarm labor input days, and -0.001 when the outcome is the household full-time farm labor share. These estimates suggest that, within the same period, households with higher mechanization levels tend to allocate less labor to farming and more labor to non-agricultural activities, which provides support for a mechanization-based interpretation of the underlying mechanism.

I further examine heterogeneity in household labor reallocation by pre-reform income, education, and land endowments to understand the distributional consequence of the reform. I present details in Appendix B.1. In summary, I find that the labor-reallocation effects are larger among households with higher pre-treatment income and a greater share of high school graduates, as reflected in larger declines in farm labor input and larger increases in nonfarm labor input. By contrast, these effects are weaker among households with larger farmland endowments. The results indicate that labor reallocation following the reform was more pronounced among households with greater economic resources and human capital, while less effective among those with stronger agricultural land endowments.

4.3 Gender Differences

To conclude the empirical analysis of mechanization and labor allocation, I examine whether the labor response to the agricultural tax reform differs by gender and the heterogeneous effects related with gender-specific characteristics.

This analysis is motivated by the conceptual framework and the discussion in the introduction, which suggest that women in rural China were more likely to remain in labor-intensive farm work before the reform because of family responsibilities, limited outside opportunities, and traditional gender norms. If mechanization substituted for manual farm labor, the reform should therefore have generated larger effects on women's labor allocation. Table 5 shows that the reduction in full-time farm is substantially larger for women than for men. In Column (1), the reform reduces the likelihood that a female worker engages in full-time farm activities by 2.0 percentage points, which is about four times as large as the male estimate in Column (2) by only 0.5 percentage points. Figure 6 presents the estimated coefficients using the corresponding event-study approach. The insignificant estimates before their treatment year implicates the parallel pre-trend.

Table 5: Gender and Heterogeneous Responses on Individual Labor Allocation

Outcome variable:	Full-time farm					
Interaction variable:			Province sex ratio at birth		Family poverty relative to neighbors	
	Female	Male	Female	Male	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
$Reform_{it}$	-0.020***	-0.005*	-0.008***	-0.004	0.002	-0.017*
	(0.003)	(0.003)	(0.003)	(0.003)	(0.005)	(0.010)
$Reform_{it} \times interaction$			-0.026***	0.002	-0.022***	0.014
			(0.004)	(0.003)	(0.007)	(0.010)
Individual FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y	Y
Observation	82,011	76,732	82,011	76,732	40,215	38,027
R-squared	0.856	0.885	0.856	0.885	0.857	0.882

Note: This table CHARLS Life History Survey restricting the sample to individuals aged 16 to 60 who are not enrolled in school in each year. Columns (1) and (2) employ specification (2) with female only and male only individuals, respectively, and do not include interaction terms. Columns (3) to (6) employ specification (2) with the corresponding interaction terms. The province sex ratio at birth is measured at the province level as the ratio of boys to girls among newborns recorded in the Fifth National Population Census. Family relative poverty is from the CHARLS questionnaire. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

The remaining columns in Table 5 provides estimation of heterogeneous effects. Column (3) and (4) examine whether this gender gap varies with the province sex ratio at birth. For women, the interaction coefficient is -0.026, while the corresponding estimate for men is close to zero and statistically insignificant. This finding indicates that the reform's effect on reducing female full-time farm is stronger in provinces with more male-biased sex ratios at birth. A more imbalanced sex ratio is often interpreted as reflecting stronger son preference and more traditional gender norms. In such settings, women are likely to be more tightly attached to agricultural and household responsibilities

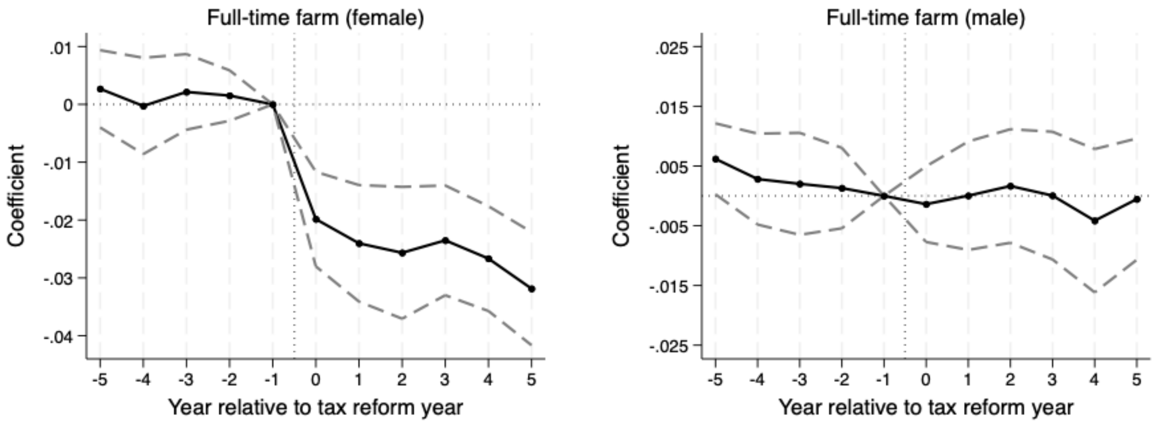


Figure 6: Event Study of Table 5

Note: This figure plots the coefficients and 95% confidence intervals of the event-study specification using data from Table 5.

before the reform.[11] Therefore, the larger post-reform decline in female farming in these areas is consistent with the view that mechanization relaxes gender-specific constraints that had previously kept women in farm work.

Finally, Column (5) and (6) examine whether the gender gap in reform responses varies with family poverty status relative to neighbors. For women, the interaction coefficient is negative and statistically significant (-0.022), while the corresponding estimate for men is positive and statistically insignificant. This finding suggests that the reform's effect on reducing female full-time farm is stronger among women from relatively poorer families, whereas no similar change emerges for men. Relative poverty may capture tighter household resource constraints and more limited outside options, but these constraints are likely to fall more heavily on women, who are more often expected to remain in farm production and household responsibilities when families face economic hardship (Tsiboe et al., 2016).[12]

[11] A better understanding is from Appendix Table B2 where I use the pre-treatment data to show the correlations between the full-time farm engagement and the sex ratio at birth. In Columns (1) and (2), I find a positive and significant relationship between these two variables for female, while small and insignificant for male, suggesting in region with more imbalanced gender preferences, female is more likely to work in agricultural sectors.

[12] Similar with the previous footnote, in Appendix Table B2 Columns (3) and (4), I find a positive correlation between full-time farm and household poverty. And the correlation is larger for female.

5 Effects of Tax Reform on Teenagers and Long-Term Outcomes

This section examines the intergenerational effects of the tax reform on teenagers who were still at school age when the reform took place. I first present the baseline results on educational attainment and other long-run outcomes by gender. I then investigate how these effects vary across household and regional characteristics.

5.1 Educational Attainment and Long-Term Benefits

This subsection examines how exposure to the tax reform during adolescence affected teenagers' later-life outcomes in education, economic status, and household labor allocation. I compare cohorts who were aged 12–15 at the time of the local reform with those aged 16–19.

Motivated by the gender differences in adult workers' responses documented in Table 5, I present the cohort DID estimates separately for girls and boys. As discussed in the conceptual framework, the reform-induced reduction in farm labor demand appears to benefit female workers more strongly. If these gains were transmitted across generations, girls should also experience larger improvements in human capital accumulation. Columns (1) to (4) of Table 6 support this prediction. For girls, exposure to the reform increases the probability of high school enrollment by 1.7 percentage points and high school completion by 1.5 percentage points. By contrast, the corresponding estimates for boys are only about 0.7 percentage points and are statistically insignificant.

Table 6: Effects of Agricultural Tax Reform on Long-Run Outcomes

Outcome variable:	High school enrollment		High school completion		Poverty status		Full-time housework	
	Female	Male	Female	Male	Female	Male	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Reform_{it}$	0.017***	0.007	0.015***	0.007	-0.007***	0.001	-0.036***	0.001
	(0.005)	(0.005)	(0.005)	(0.005)	(0.001)	(0.001)	(0.006)	(0.001)
County FE	Y	Y	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y	Y	Y	Y
Observation	110,661	109,290	98,071	95,590	110,661	109,290	56,534	53,145
R-squared	0.488	0.556	0.539	0.623	0.026	0.026	0.087	0.082

Note: This table employs specification (3) to compare teenagers aged 12–15 with those aged 16–19 in the tax reform year, using data from the 2010 Census Sample Survey. Poverty status is defined as receiving government poverty assistance in 2010. Full-time housework indicates engagement in full-time household work; individuals who were unable to work are excluded in Column (7) and (8). $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

The educational gains for girls are also accompanied by meaningful improvements in later-life socioeconomic outcomes, motivated by the literature (Zhang and Meng, 2025; Shi, 2026). Columns (5) and (6) show that exposure to the reform reduces the probability that a girl later receives government poverty assistance by 0.7 percentage points, while the corresponding estimate for boys is close to zero and statistically insignificant. Relative to the baseline mean of 1.1 percent for girls, this effect implies a decline of about one-half. Columns (7) and (8) reveal a similar gendered effects for household labor allocation in adulthood. For girls exposed to the reform before the high school margin, the probability of engaging in full-time housework in 2010 falls by 3.6 percentage points, whereas the estimate for boys is again small and insignificant. Given the baseline mean for girls, this corresponds to roughly a one-third reduction.

The corresponding event-study estimates are plotted in Figure 7. The insignificant pre-treatment estimates suggest the parallel trend in the pre-treatment periods. And the post-treatment estimates are close to the DID estimates, supporting the robustness of the DID estimates.

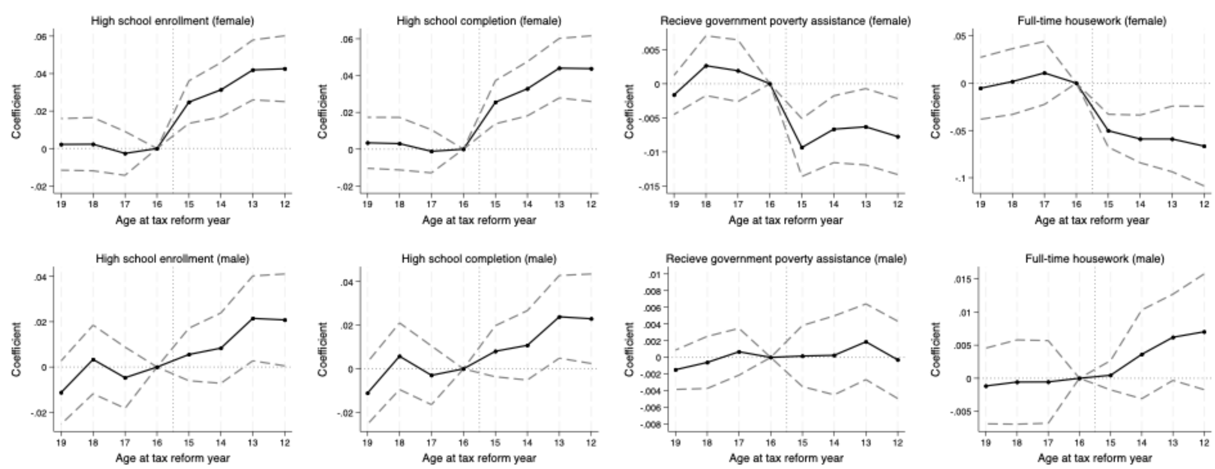


Figure 7: Event Study of Table 6

Note: This figure plots the coefficients and 95% confidence intervals of the event-study specification using data from Table 6.

5.2 Mechanisms and Heterogeneity Analysis

This subsection examines how the long-run gains for girls vary across different household and regional environments. Table 7 extends the baseline cohort DID specification by interacting the treatment indicator with four variable: sibling status, mother's nonfarm work experience, the provincial sex ratio at birth, and the provincial share of the population engaged in farming.

Table 7: Mechanism Analysis and Heterogeneous Responses on High School Completion

Outcome variable:	High school completion							
Interaction variable:	Have siblings		Mother never worked in nonfarm sectors		Province sex ratio at birth		Farm population share	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)	Female (7)	Male (8)
$Reform_{it}$	0.061***	-0.032	0.096***	-0.037	0.010*	0.006	-0.010	-0.009
	(0.020)	(0.037)	(0.039)	(0.035)	(0.006)	(0.006)	(0.011)	(0.010)
$Reform_{it} \times interaction$	0.053***	-0.005	0.015***	0.008	0.012**	0.003	0.038**	0.024*
	(0.021)	(0.017)	(0.004)	(0.044)	(0.006)	(0.005)	(0.015)	(0.014)
County FE	Y	Y	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y	Y	Y	Y
Observation	9,732	11,326	9,732	11,326	110,661	109,290	110,661	109,290
R-squared	0.302	0.210	0.303	0.210	0.488	0.556	0.488	0.556

Note: This table employs specification (3) to compare teenagers aged 12–15 with those aged 16–19 in the tax reform year. Column (1) to (4) use CHIP waves of 2013 and 2018. Column (5) to (8) uses 2010 Census Sample Survey. Mother never worked in nonfarm sectors indicates the teenager's mother only worked in agricultural sectors or did not work. The sex ratio at birth and farm population share are on province level and from the Fifth National Population Census. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

Column (1) to (4) of Table 7 provide evidence consistent with the two mechanisms discussed in the conceptual framework: the income effect of the reform and the change in parents' expectations about their children's (daughter's) future work trajectories. The first two columns use sibling status to capture the extent to which household schooling decisions are constrained by financial resources, while Column (3) and (4) use mothers' nonfarm work history to proxy for the family's prior exposure to non-agricultural opportunities and, relatedly, parents' expectations about children's future employment choices.

Columns (1) and (2) support the income-effect channel. For girls, the coefficient on the reform indicator is 0.061, implying that even among those from one-child households, exposure to the reform significantly increases high school completion. More importantly, the interaction term with having siblings is positive and statistically significant at 0.053, indicating larger gains for girls from multi-child families. This result is consistent with the idea that schooling investments are more likely to be limited by financial constraints in households with multiple children, where daughters may face a higher risk of giving up educational opportunities when family resources are limited.

The tax relief generated by the reform appears to relax these constraints more strongly in such families. For boys, by contrast, neither the reform effect nor the interaction term is statistically significant.

Columns (3) and (4) turn to the expectations channel. Here the interaction variable indicates that the teenager's mother never worked in nonfarm sectors, which reflects a household environment with more limited exposure to non-agricultural employment and potentially more traditional expectations about daughters' future roles. For girls, the estimated reform effect is 0.096, and the interaction term is an additional 0.015. These estimates imply substantially larger gains in high school completion for girls whose mothers lacked nonfarm work experience. This finding is consistent with the view that the reform altered parents' perceptions of the returns to girls' education by making nonfarm opportunities more relevant and attainable. Again, the corresponding coefficients for boys appear small and statistically insignificant.

Additional understanding of the estimated effects is facilitated by heterogeneous analysis. Columns (5) and (6) show that the reform's effect on girls' high school completion is larger in provinces with more male-biased sex ratios at birth. For girls, both the reform coefficient and the interaction term are positive and significant, while for boys both estimates are small and insignificant. This finding suggests that the educational gains for girls are stronger in places with stronger son preference, which is also consistent with the estimates in Table 5 where the treatment effects on labor allocation are larger among female workers in regions with stronger son preferences

Columns (7) and (8) show that the reform's effect is also stronger in areas with a higher farm population share. The interaction term is positive and significant for girls, and smaller and only weakly significant for boys. Since in more agriculture-dependent areas the reform likely produced larger changes in household labor allocation and in the perceived value of education relative to farm work, stronger local agricultural dependence may amplify the educational effects of the reform.

6 Robustness Checks

6.1 Liquidity Constraints

An alternative explanation is that the tax cut relaxes households' liquidity constraints. As emphasized in the literature, rural households often cannot borrow freely against future income and therefore depend heavily on current cash flow to finance the upfront costs of migration or commuting for nonfarm work (Bryan et al., 2014; Hidrobo et al., 2022). From this perspective, lower agricultural taxes may facilitate movement into nonfarm employment by directly easing financial constraints, rather than through the labor-substituting effect of farm mechanization.

To assess the liquidity-constraint explanation, Table C1 Column (1) interacts the reform indicator with whether the household borrowed money as a proxy for household liquidity constraints, while Column (2) interacts the reform indicator with the distance to the nearest train station to proxy for nonfarm work costs. In both cases, the interaction terms are statistically insignificant.

Column (3) includes household borrowing directly as a control variable, and the reform coefficient remains positive and significant. As a comparison, Column (4) includes the farm production machinery value per unit of farmland as a control variable. The reform coefficient becomes smaller, while the machinery variable is positive and statistically significant. These results provide little evidence that the reform's effect on nonfarm labor supply is primarily driven by the relaxation of liquidity constraints or lower mobility costs.

6.2 Partial Effects on Control Cohorts

Another concern is that some teenagers in the control group may still be partially affected by the tax reform, due to the potential decline in high school dropout and the deferral of high school entry.

To address the first concern, I have already shown in the baseline analysis that using either high school enrollment or high school completion as the outcome produces very similar estimates. Here I provide additional evidence by examining high school dropout directly. Figure C1 plots dropout rates by cohort for the full sample and shows no visible break over cohorts. Consistent with this result, Table C2 reports cohort DID estimates using an indicator for dropping out of high school as the outcome variable. The estimated coefficients are small and statistically insignificant, providing no evidence that the reform affected dropout behavior. This finding suggests that the reform mainly influenced educational attainment through the decision to enter high school rather than through changes in dropout after enrollment.

To address the second concern related to students who defer entry into high school, I re-estimate the cohort DID specification by excluding teenagers aged 16 and 17 at the year of the reform from the control group. This alternative specification compares teenagers aged 12–15 with those aged

18–19. As reported in Table C3, the estimated effect remains very similar to the baseline result. This finding suggests that the baseline results are robust to excluding the cohorts most likely to contain deferred students, and that this partial exposure is unlikely to bias the baseline estimates.

6.3 Supply of Educational Resources

The documented improvements in high school attainment may reflect an expansion in educational resources rather than a change in household demand for education. To examine this possibility, I supplement the individual-level analysis with province-level panel data from the annual statistical yearbooks for 1996–2008 and estimate an alternative staggered DID specification on education supply outcomes. Because county-level measures of educational resources are not available, treatment in this specification is defined as the first year in which the reform had been fully implemented in all counties within a province, and all subsequent years. The regressions include province and year fixed effects as well as province-specific linear time trends.

Table C4 reports the results for three measures of high school education supply: fiscal education expenditure, the number of rural high schools, and the number of registered rural high school teachers. Across all three outcomes, the estimated coefficients are small and statistically insignificant, providing no evidence that the baseline estimates are a result of increased local supply of educational resources.

6.4 Middle School Completion

The estimated gender gap in high school completion may also reflect differences in middle school completion. The cohort DID design implicitly assumes that, under the nine-year compulsory education system, teenagers aged 15 and above should already have completed middle school. In rural areas, however, compulsory schooling may have been enforced less strictly, even though enforcement likely improved over time. While such gradual changes should be absorbed by the fixed effects and county-specific trends, I provide additional evidence to assess this possibility directly.

Table C5 repeats the baseline cohort DID specification with middle school completion as the outcome variable. If the estimated gender gap in high school completion were driven by differences in middle school attainment, we would expect to observe corresponding differences at this earlier margin as well. The estimated coefficients, however, are very small, suggesting little difference in middle school completion between the treatment and control cohorts and making it unlikely that the baseline findings on high school attainment are driven by pre-existing differences at the middle school stage.

6.5 Migration

The estimated effects of the agricultural tax reform may also be affected by changes in residence. If some rural households moved to counties with better nonfarm job opportunities or richer educational resources around the reform period, the DID estimates could partly reflect relocation decisions.

Figure C5 shows the share of rural households that changed their registered place of residence in my sample. The share is very small throughout the sample period, indicating that permanent relocation is rare in the panel data. This descriptive evidence suggests that large-scale sorting of rural households across counties is unlikely to drive the main results.

I address this concern by re-estimating both the short-run and long-run DID specifications after excluding all households that ever changed residence between 1996 and 2008. Tables C6 and C7 report the short-run effects on labor outcomes in this restricted sample, and the estimates remain very similar to the baseline results. Table C8 repeats the cohort DID analysis for the long-run outcomes using the non-migrant sample. The estimated effects on girls' high school completion, receipt of poverty-related government assistance, and engagement in full-time housework are again almost identical to those in Table 6, suggesting no evidence of the confounding effects of permanent relocation.

6.6 Confounding Policy

China's family planning policy, especially the one-child policy, may also confound the estimated effects. As the policy became more stringent, the number of children in rural households declined, as shown in Figure C8. If this reduction in sibship size coincided with the agricultural tax reform, the increase in high school completion documented above could partly reflect parents reallocating resources toward fewer children.

To assess this possibility, I test whether exposure to the agricultural tax reform is systematically associated with changes in family structure. Specifically, I re-estimate the baseline cohort DID specification using the number of siblings as the outcome, separately for girls and boys. Table C9 reports the results. The estimated coefficients are positive but only marginally significant at 0.023 for girls and 0.012 for boys. This result suggests that treated cohorts are, if anything, slightly more likely to have siblings than the control cohorts. In particular, there is no evidence that teenagers exposed to the tax reform were more likely to be only children once fixed effects are taken into account.

6.7 Staggered DID and TWFE estimates

The TWFE specification combines comparisons across groups treated at different times, and some of these comparisons may receive negative weights when treatment effects are heterogeneous across groups or over time (de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021). I follow Callaway and Sant'Anna (2021) to address this concern. Specifically, I group counties by their first treatment year and compare each treated group with counties that are not yet treated in the same year. The resulting group-time treatment effects are then aggregated to obtain the average treatment effect on the treated. This approach avoids comparisons between later-treated groups and already-treated groups.

Appendix Table C10 reports the aggregated treatment effects on household labor allocation and farm machinery investment. The estimates are similar to the staggered DID results in Tables 2 and 3, although somewhat smaller in magnitude, which supports the robustness of the baseline staggered DID estimates.

7 Conclusions

This paper investigates how the abolition of China's agricultural tax reshaped rural labor allocation, enhanced agricultural productivity, and generated long-lasting intergenerational consequences. Leveraging the staggered county-level roll-out of the reform from 2000 to 2006, I provide causal evidence on how reductions in the effective cost of production triggered substantial investment in agricultural machinery, reduced reliance on manual labor, and induced workers to reallocate toward higher-productivity nonfarm sectors. These production-side adjustments offer micro-level insights into the mechanisms underlying sectoral misallocation.

Beyond short-run labor responses, the reform produced persistent gains for teenagers who were exposed during critical schooling margins. Using a cohort difference-in-differences strategy, I find that cohorts exposed to reform before entering high school experienced higher high school enrollment and completion likelihood, lower poverty incidence in adulthood, and reduced engagement in full-time housework. These findings highlight how productivity-enhancing agricultural reforms can relax both financial and time-use constraints within households, ultimately encouraging investments in children's human capital and influencing their long-run socioeconomic trajectories.

The analysis also uncovers substantial gender heterogeneity. Because women were disproportionately engaged in manual farm work prior to the reform, the labor-saving effects of mechanization freed up female labor to a greater extent. This differential response was larger in regions with stronger son preference, lower pre-reform mechanization, or higher agricultural dependence. These gendered shifts in labor allocation translated into correspondingly larger intergenerational benefits for girls, underscoring the potential for productivity shocks to alleviate gender-specific constraints in agrarian economies.

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A Descriptive Statistics

A.1 Descriptive Evidence on Conceptualization

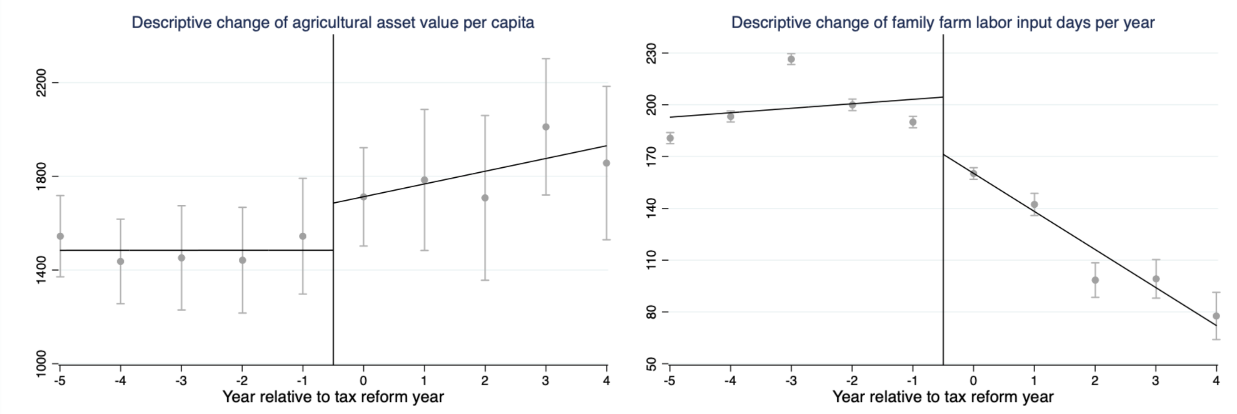

Figure A1: Changes in Capital and Labor Inputs in Farming Production

Note: Data is from NFPS

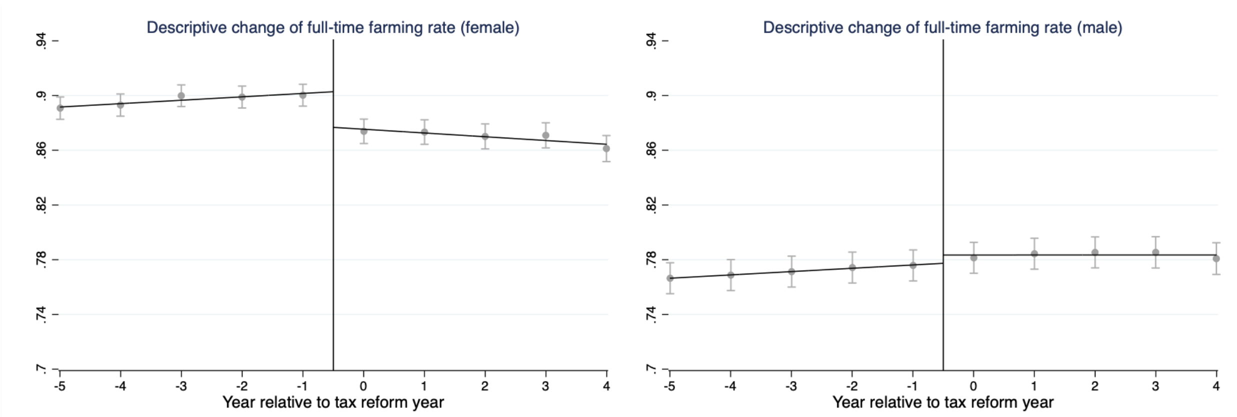

Figure A2: Gender Differences in Labor Reallocation

Note: Data is from CHARLS Life History Survey

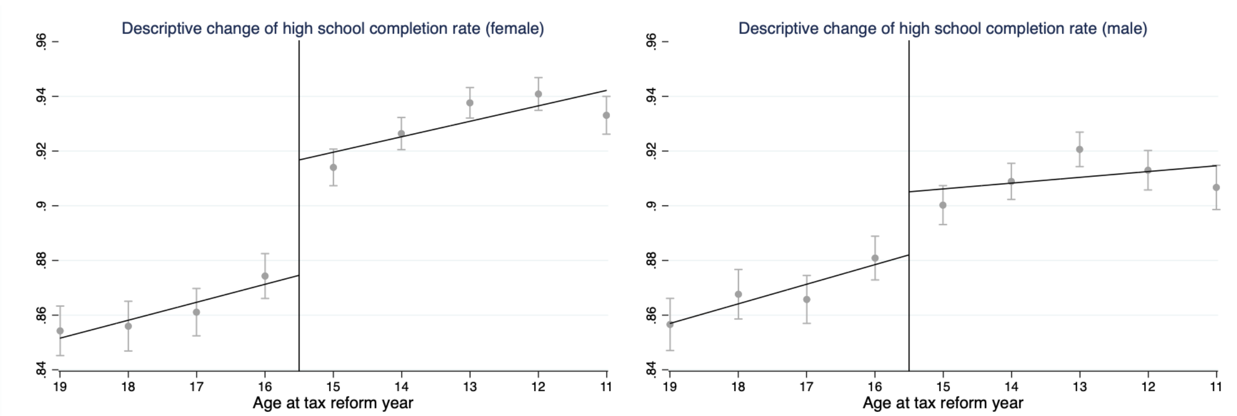

Figure A3: Gender Differences in High School Completion

Note: Data is from 2010 Census Sample Survey

A.2 Additional Descriptive Statistics

Table A1: Descriptive statistics (pre-treatment) and data sources

Data level	Variable	Gender	Observation	Mean	Std. dev.
Household level	Agricultural tax per capita		25,407	223.308	469.765
	Farming income per capita		25,407	1,004.986	1,366.452
	Agricultural asset value (CNY) per capita		25,407	1,263.873	7,997.588
Individual level	Labor force participation	Female	30,757	0.831	0.375
		Male	27,191	0.894	0.308
	Full-time farm given labor force participants	Female	25,559	0.891	0.312
		Male	24,275	0.758	0.428
	High school completion	Female	23,977	0.862	0.345
		Male	22,718	0.868	0.338
	Poverty status	Female	23,977	0.011	0.104
		Male	22,718	0.003	0.058
	Full-time housework (excluding illness and in school)	Female	20,728	0.114	0.318
		Male	19,057	0.003	0.051
	Number of siblings	Female	3,412	2.340	3.406
		Male	4,026	1.373	1.044
	Mother is unemployed or only farm	Female	3436	0.195	0.396
		Male	4,042	0.131	0.338
	Feel poor than neighbors		24,597	0.848	0.36
Province level	Sex ratio at birth (girl = 100)		31	119.604	11.351
	Full-time farm share		31	0.751	0.161

Note: This table reports the descriptive statistics. It includes the variables one year before the treatment of each county. Observation is the total number of households/individuals used in the full sample.

B Heterogeneity

B.1 Heterogeneous Effects

The baseline estimates show that the agricultural tax reform reduced household labor input in farming and increased labor allocation toward non-agricultural activities. In this subsection, I examine whether these responses differ across households with different economic and demographic characteristics to assess the distributional implications of the reform. In particular, I focus on three pre-treatment household characteristics: income, education, and land endowments. These variables capture key dimensions along which households may differ in their ability to respond to the reform, including economic resources, human capital, and the extent of reliance on agricultural production.

Appendix Table B1 reports the heterogeneous treatment effects on household labor allocation by adding interaction terms to the baseline DID specifications. In Columns (1) and (2), I find that the decline in household farm labor input days is larger among households with higher pre-treatment income, with an interaction coefficient of -0.049, while the increase in household nonfarm labor input days is also more pronounced among these households, with an interaction coefficient of 0.254. These results suggest that the reform induced stronger labor reallocation among relatively better-off households.

Columns (3) and (4) show heterogeneity by the share of high school graduates within the household. The reform is expected to have stronger labor-reallocation effects in households with higher human capital, since more educated members may be better able to adjust to technological change and transition into non-agricultural work. The results are in line with this expectation. Households with a higher share of high school graduates experience a larger decline in farm labor input days (-0.249 for interaction coefficient) and a larger increase in nonfarm labor input days (1.785 for interaction coefficient). These findings suggest that higher human capital strengthened the household's ability to reallocate labor away from farming in response to the reform.

Columns (5) and (6) focus on heterogeneity by household farmland per capita. The effect of the reform is expected to be less pronounced for households with larger land endowments, as these households are more heavily engaged in agricultural production and thus have a stronger incentive to retain labor on the farm. The estimation results show that for households with more farmland per capita, the reduction in household farm labor input days is more limited (0.033 for interaction

Table B1: Heterogeneous Responses on Household Labor Allocation

Outcome variable:	Farm	Nonfarm	Farm	Nonfarm	Farm	Nonfarm
			labor input days (log)			
Interaction variable:	Household annual income per capita		Share of household high school graduates		Household farmland per capita	
	(1)	(2)	(3)	(4)	(5)	(6)
$Reform_{it}$	-0.108**	0.0004	-0.124**	-0.086	-0.265***	0.629***
	(0.045)	(0.2103)	(0.059)	(0.212)	(0.061)	(0.217)
$Reform_{it} \times interaction$	-0.049***	0.254***	-0.249***	1.785***	0.033**	-0.057**
	(0.010)	(0.041)	(0.045)	(0.199)	(0.014)	(0.022)
Household FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y	Y
Observation	25,407	25,407	25,407	25,407	25,407	25,407
R-squared	0.560	0.491	0.559	0.492	0.560	0.491

Note: This table uses NFPS household data and employs specification (1) with interaction terms. The outcome variable in Column (1), (3), and (5) is the household farm labor input days per year in log form. The outcome variable in Column (2), (4), and (6) is the household nonfarm labor input days per year in log form. The interaction variable in Column (1) and (2) is household total income per capita (in thousand CNY). It in Column (3) and (4) is share of high school graduates of each household. In Column (5) and (6) it is the area of household farmland per capita. All the interaction variables are measured in pre-treatment average and thus time-invariant. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

coefficient), and the expansion in household nonfarm labor input days is also less evident (-0.057 for interaction coefficient). This finding suggests that a larger land base reduced the extent to which households reallocated labor away from farming after the reform.

B.2 Female Heterogeneity and Norms

Table B2: Labor Division and Norms

Outcome variable:	Full-time farm			
Independent variable:	Province sex ratio at birth		Family poverty relative to neighbors	
	Female	Male	Female	Male
	(1)	(2)	(3)	(4)
Independent variable	0.048**	-0.031	0.176***	0.115**
	(0.021)	(0.019)	(0.029)	(0.023)
Year FE	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y
Observation	25,559	24,275	12,570	12,010
R-squared	0.002	0.003	0.020	0.020

Note: This table reports the correlations between labor allocation and gender norm using the pre-treatment sample. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

C Robustness

C.1 Liquidity Constraints

Table C1: Liquidity Constraints

Outcome variable:	Household nonfarm labor input days (log)			
Interaction or control variable:	Borrow money (1)	Distance to train station (2)	Borrow money (3)	Dynamic farm production machinery value per farmland (4)
$Reform_{it}$	0.514**	-0.817	0.528**	0.477**
	(0.208)	(3.016)	(0.207)	(0.195)
$Reform_{it} \times interaction$	0.127	0.278		
	(0.138)	(0.616)		
$Borrow_{it}$			-0.133*	
			(0.068)	
$Machinery_{it}$				0.0001***
				(0.0000)
Household FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y
Observation	25,407	25,407	25,407	25,407
R-squared	0.491	0.491	0.491	0.492

Note: This table uses if the household borrows money and the distance between housing to the nearest train station as proxies to the liquidity constraints. Columns (1) and (2) include the two variables as interaction terms. Column (3) controls for the indicator of money borrow. Column (4) controls for the machinery adoption as a reference. $^{*} : p < 0.1$; $^{**} : p < 0.05$; $^{***} : p < 0.01$. Standard errors in parentheses are clustered at the county level.

C.2 Partial Effects on Other Cohorts

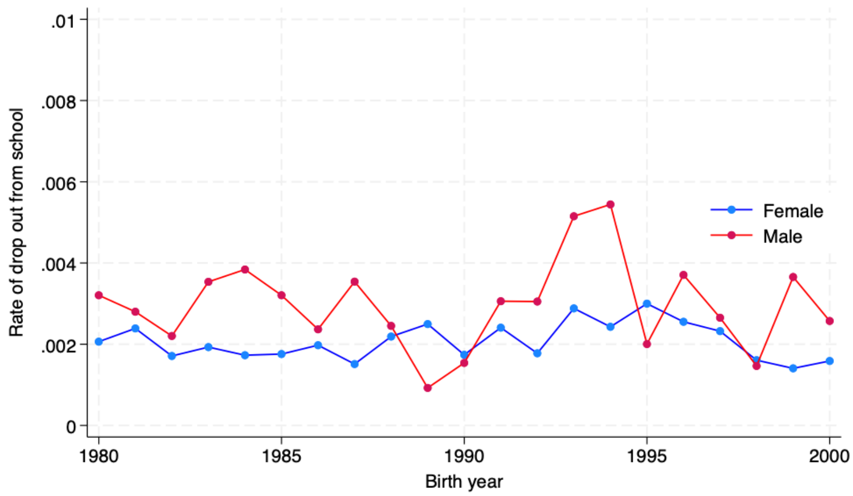

Figure C1: Changes in High School Drop Out Rate

Note: Data is from 2010 Census Sample Survey

Table C2: Effects of Tax Reform on High School Drop Out

	Female	Male
	(1)	(2)
$Reform_{it}$	0.000	-0.002*
	(0.001)	(0.001)
County FE	Y	Y
Cohort FE	Y	Y
County-specific trend	Y	Y
Observation	110,495	109,143
R-squared	0.031	0.030

Note: This table employs specification (3) using 2010 Census Sample Survey. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

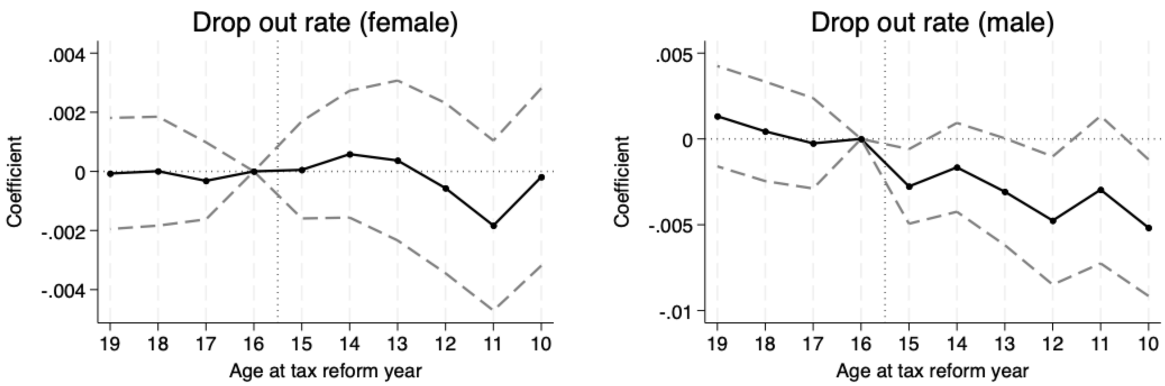

Figure C2: Event Study of Table C2

Table C3: Drop Cohorts of 16 and 17 Years Old at Reform Years

	High school completion		Poverty status		Full-time housework	
	Female	Male	Female	Male	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
$Reform_{it}$	0.019**	0.001	-0.001	-0.001	-0.016**	0.001
	(0.008)	(0.008)	(0.001)	(0.001)	(0.008)	(0.001)
County FE	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y	Y
Observation	98,250	97,169	98,250	97,169	46,640	43,878
R-squared	0.513	0.580	0.030	0.030	0.098	0.097

Note: This table repeat Table 6 while drop individuals of 16 or 17 years old at reform years. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

C.3 Supply change of high school education

Table C4: Change of Local Education Resources

	Education expenditure (log)	Number of high school	Number of high school teacher
	(1)	(2)	(3)
$Reform_{it}$	0.007	-2.560	-4391.211
	(0.020)	(7.837)	(3421.743)
Province FE	Y	Y	Y
Year FE	Y	Y	Y
Province-specific trend	Y	Y	Y
Observation	374	389	389
R-squared	0.993	0.989	0.992

Note: This table employs specification (1) to examine the change of local education facilities and teacher numbers after tax reform. Data is on province level from annual statistical books from 1996 to 2008. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the province level.

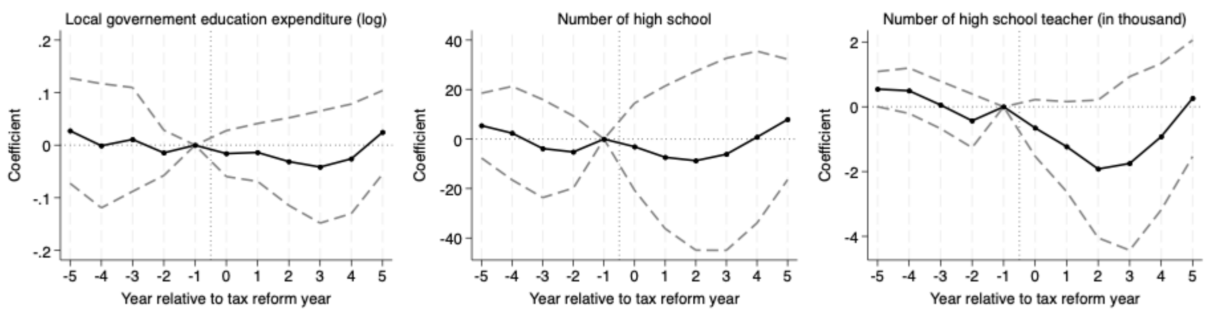

Figure C3: Event Study of Table C4

C.4 Middle School Completion

Table C5: Middle School Completion

Outcome variable:	Middle school completion	
	Female	Male
	(1)	(2)
$Reform_{it}$	0.003*	0.002
	(0.002)	(0.002)
County FE	Y	Y
Cohort FE	Y	Y
County-specific trend	Y	Y
Observation	110,661	109,290
R-squared	0.718	0.746

Note: This table employs specification (3) using 2010 Census Sample Survey. The outcome variable is middle school completion. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

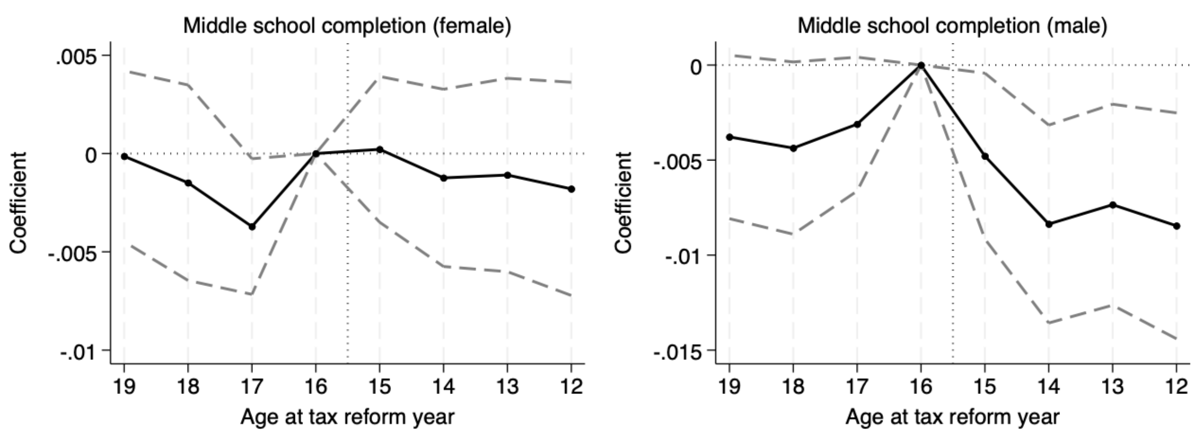

Figure C4: Event Study of Table C5

C.5 Change of Residence Place

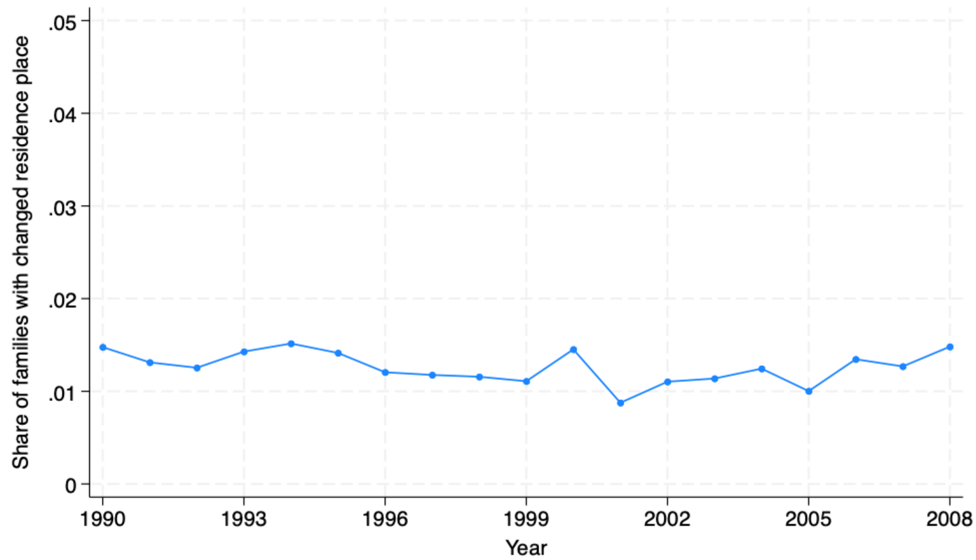

Figure C5: Change in Migration Rate

Table C6: Drop households who changed residence place

	Labor force participation (1)	Full-time farm over workers (2)
$Reform_{it}$	0.002	-0.011***
	(0.003)	(0.002)
Individual FE	Y	Y
Year FE	Y	Y
County-specific trend	Y	Y
Observation	168,229	142,258
R-squared	0.651	0.889

Note: This table employs specification (2) and CHARLS Life History Survey. I drop households who migrated between 1996 and 2008. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C7: Drop households who changed residence place

	Labor force participation		Full-time farm	
	Female	Male	Female	Male
	(1)	(2)	(3)	(4)
$Reform_{it}$	0.004	0.000	-0.019***	-0.003
	(0.005)	(0.004)	(0.003)	(0.002)
Individual FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y
Observation	92,616	75,613	75,558	66,700
R-squared	0.686	0.586	0.872	0.900

Note: This table employs specification (2) and CHARLS Life History Survey. I drop households who migrated between 1996 and 2008. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C8: Drop households who changed residence place

	High school completion		Poverty status		Full-time housework	
	Female	Male	Female	Male	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
$Reform_{it}$	0.016**	0.010	-0.006***	0.001	-0.034***	0.002*
	(0.007)	(0.008)	(0.002)	(0.003)	(0.009)	(0.001)
County FE	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y	Y	Y
Observation	56,602	58,894	56,602	58,894	27,882	27,961
R-squared	0.539	0.589	0.044	0.039	0.127	0.137

Note: This table employs specification (3) and 2010 Census Sample Survey. I drop households who migrated between 1996 and 2008. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

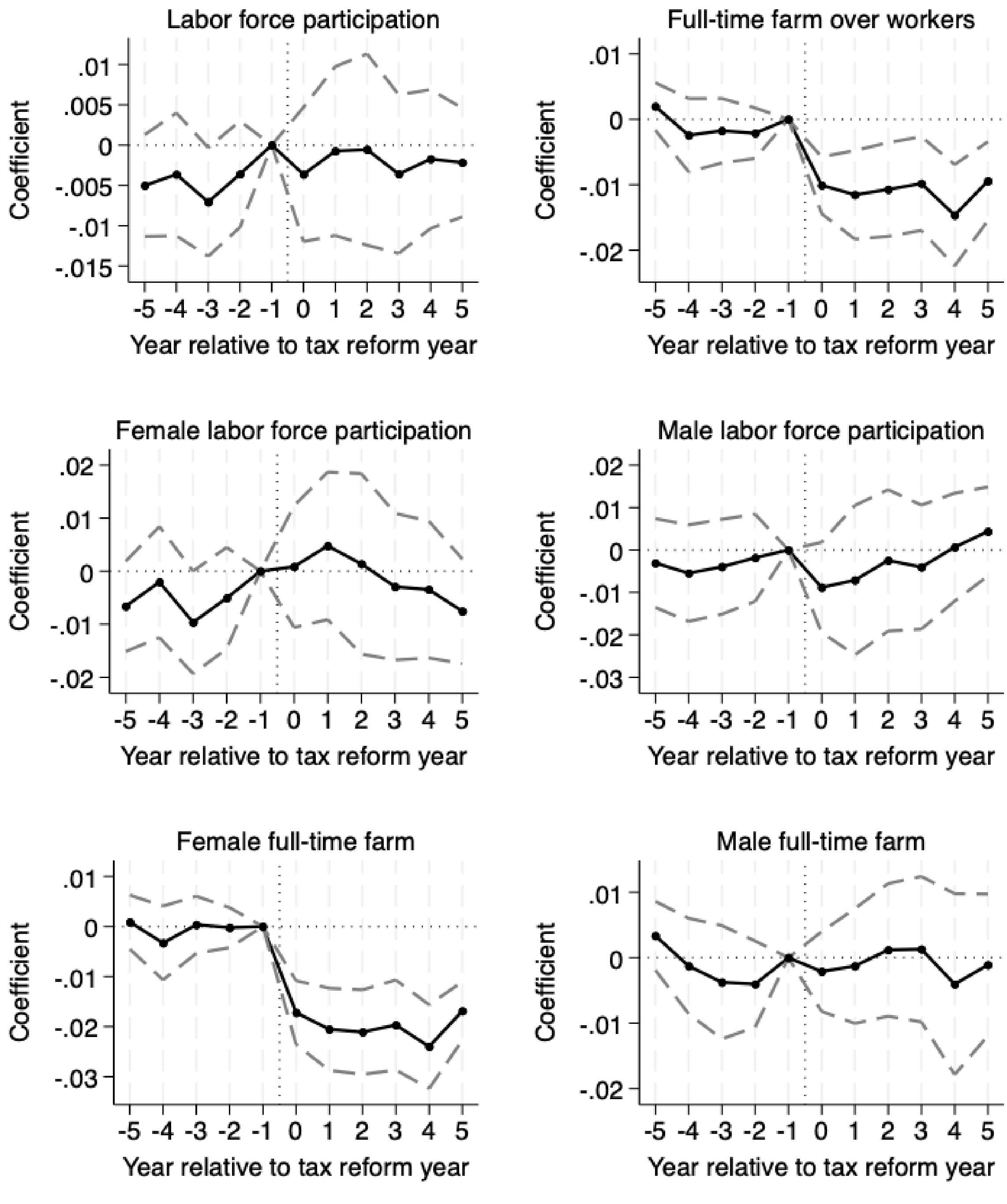

Figure C6: Event Study of Table C6 and C7

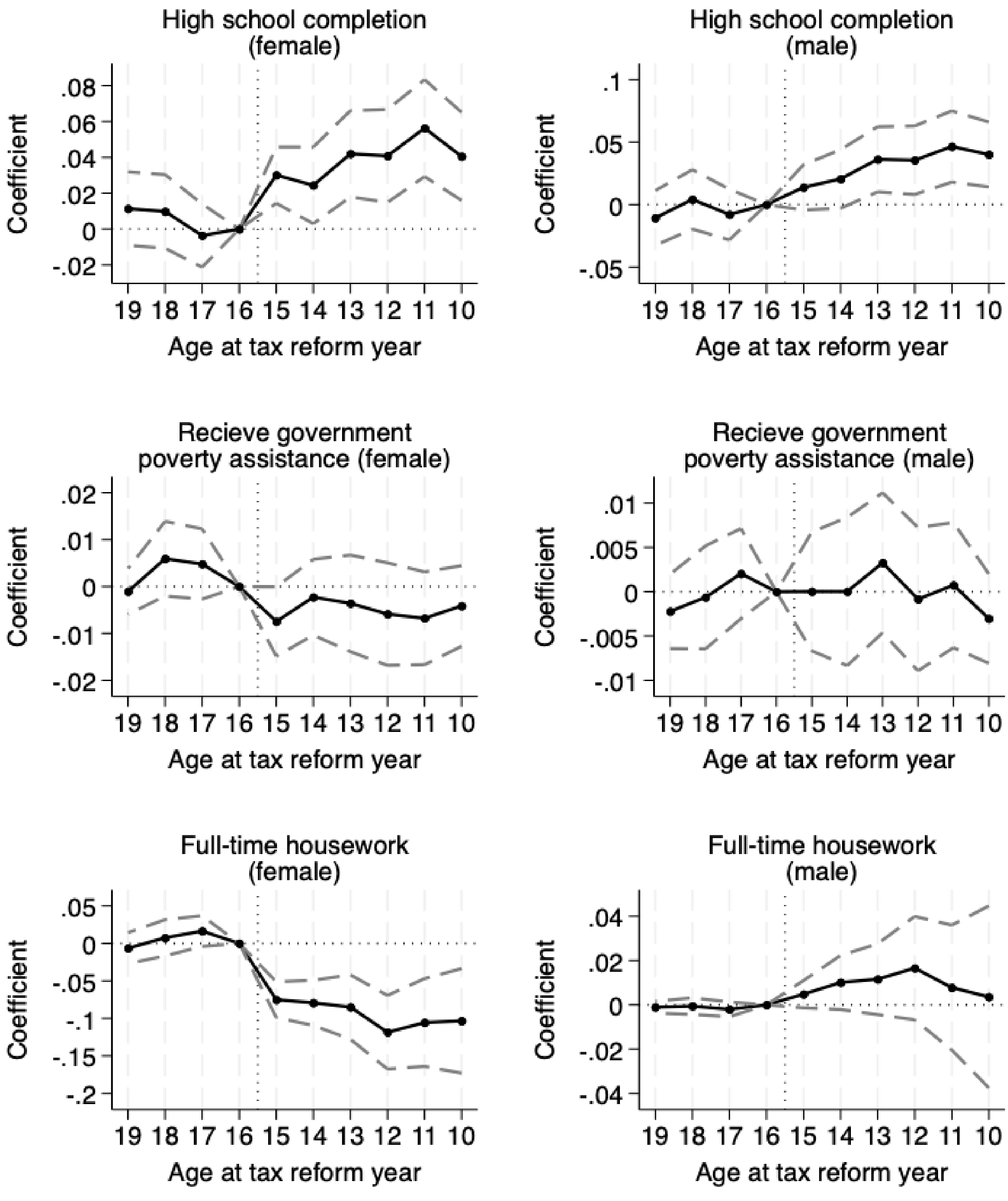

Figure C7: Event Study of Table C8

C.6 Confounding Policy

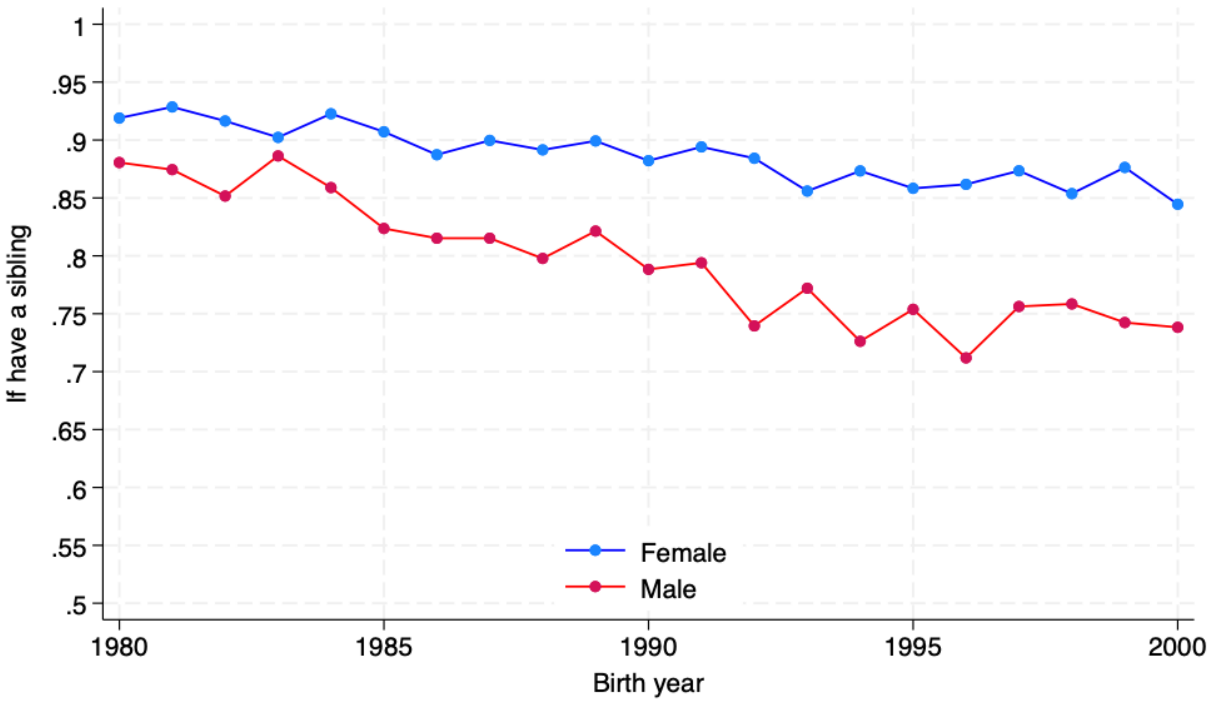

Figure C8: Changes in Having Siblings in Families

Table C9: Effects of OCP

	Female (1)	Male (2)
$Reform_{it}$	0.023*	0.012*
	(0.012)	(0.007)
County FE	Y	Y
Cohort FE	Y	Y
County-specific trend	Y	Y
Observation	9,698	11,293
R-squared	0.046	0.321

Note: This table employs specification (3) using CHIP waves of 2013 and 2018. The outcome variable is the number of siblings. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.

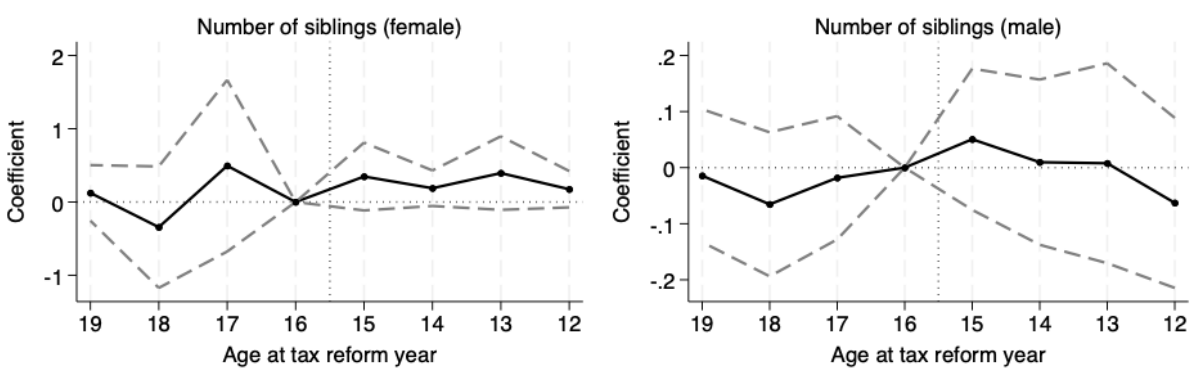

Figure C9: Event Study of Table C9

C.7 Staggered DID

Table C10: Robustness Checks for Staggered DID Specification

Outcome variable:	Household farm labor input days (log)	Household nonfarm labor input days (log)	Farm production machinery value per farmland (log)	Farm processing machinery value per farmland (log)
	(1)	(2)	(3)	(4)
$Reform_{it}$	-0.147***	0.417**	0.249**	0.141***
	(0.036)	(0.201)	(0.100)	(0.036)
Household FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County-specific trend	Y	Y	Y	Y
Observation	25,407	25,407	25,407	25,407

Note: This table uses NFPS household data and employs the estimator proposed by Callaway and Sant'Anna (2021). The estimates are derived by aggregating group-time treatment effects. The control groups are the not-yet-treated counties. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses are clustered at the county level.